

Verdier-Compatible Kernel Skeletonization under Anchored Semistable Refinement

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Abstract

For four anchored semistable models of $xy = t^4$, we prove a one-sided kernel descent theorem with geometric rational unipotent nearby coefficients and complete inverse-image supports. The actual counit $U_b \star T(P) \rightarrow P \star U_a$ is invertible on a specified generated kernel source. This avoids identifying the pullback of a coarse diagonal with a fine diagonal, and makes proper product direct image strongly compatible with convolution. Graph and rectangular kernels provide the finite geometric control needed to prove boundedness and the counit's invertibility. The resulting anchored localization admits a Morita realization preserving convolution and units, with actual geometric Verdier mates and coherent support base change, localization boundaries and refinement comparisons. The cellular target is constructed independently from strata and links. An open stratum example distinguishes geometric kernel maps from the larger Hochschild mapping objects in the Morita target; the realization is not fully faithful. A separate appendix proves a local point-valued completed-root realization and its carrier pairing, with separate ambient graph calculations and two geometric coarse rectangular tests. Another appendix proves an obstruction to ordinary stalk/pullback gluing of distribution diagonals. The exceptional mate retains a noninvertible nodal dualizing complex. A global completed root kernel calculus is a separate research problem; the main theorem uses only the nearby-enhanced source and is independent of that problem.

1 One-sided descent and the main results

The central comparison is a map of actual kernels, not only of their coefficient actions. For a refinement, let U_a denote its selected pullback graph and T proper direct image on the product of supports. We prove $U_b \star T(P) \simeq P \star U_a$. The comparison is one-sided: it does not identify a fibre-relation kernel with a fine diagonal. Its counit description is what makes the ensuing convolution and Verdier comparisons compatible.

Let R be a complete strictly henselian trait with algebraically closed residue field k , uniformizer t , and $\ell \neq \text{char } k$. Fix the coordinates and compatible base roots. For $xy = t^4$ fix $B = \{2\}$ and the index

$$\mathcal{J}_B = \{B, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Let X_J be the actual integer-cut models of [2], and $u_{JI} : X_J \rightarrow X_I$ their proper contractions for $I \subset J$. All cut inclusions in this paper allow equality. The coarse exceptional curve E_B is $\mathbf{P}_k^1$, stratified by $U = \mathbf{G}_{m,k}$ and its endpoints q_0, q_∞ . For a locally closed union S of coarse strata put $S_J = u_{JB}^{-1}(S)$, with its full scheme-theoretic inverse image. Products of supports in this paper are over k , not over R .

The coefficient algebra H_{S_J} is the restriction of the actual geometric nearby image of the constant coefficient, with its algebra structure retained. Base Galois action is forgotten,

not replaced by inertia invariants. The ambient module category is the fibrewise-cartesian category $\mathcal{A}_{J|B}(S_J)$ of [2]; attachments between distinct coarse strata remain arbitrary. The smaller unit-generated category $\mathcal{C}_J(S_J)$ is not substituted for this ambient category.

The geometric inputs from [2] are its branch-algebra comparison, the algebra-unit comparison $H \rightarrow Ru_*H$ on complete coarse supports, and its “Comparison of the ambient gluing maps” lemma. The latter compares arbitrary coarse attachments under module pull-back; it does not assert kernel descent on products. We use also the actual composition of its integer-cut contractions. The new product comparison, convolution closure and their joint Verdier compatibility are proved here. Thus these earlier geometric results are inputs, whereas preservation by a generated kernel and its product descent are conclusions, not defining conditions.

1.1 Framework and input contract

In the main text preceding the appendices, categories are $\mathbf{Q}_\ell$ -linear stable infinity categories. The appendices instead use their specified $\Lambda_r = \mathbf{Z}/\ell^r$ -linear and integral categories; rationalization there occurs only after the stated adic recovery. In either setting “full” retains mapping spectra. We use a category enriched in Cat_∞ , equivalently its enriched nerve as an $(\infty, 2)$ -category. Its objects are decorated supports (J, S) ; its Hom infinity categories are the bimodule-kernel categories of Definition 2.5. Kernels are 1-morphisms and kernel maps are 2-morphisms. For each finite string the composition functor is the multiple-product shriek direct image of the relative multibar tensor. Substitution is the Fubini map of these diagrams; the empty string is Δ_*H . Thus the enriched composition data are specified by the bar construction, not by an unspecified choice of associators in a triangulated category. Restrict to the full generated Hom categories; closure is proved below. We call this locally stable enriched category $\mathbb{K}_!$. The notation in the main theorem denotes this category with its decorations.

The supported category $\mathbb{K}_*$ has objects a^D and Hom categories $\mathbb{K}_!(a, b)^{op}$, realized by the actual bounded Verdier labels. Its composition is the dual of shriek composition. Here co takes the opposite of every Hom category, leaving objects and 1-morphism directions fixed. Both geometric duality and cellular duality are enriched functors out of this co category. The cellular target is the enriched category of the specified algebras and right-target-perfect bimodules.

We invert refinement *1-morphisms*. Precisely, a localization is an enriched functor $L : \mathbb{K}_! \rightarrow \mathbb{L}$ for which, for every locally stable enriched category $\mathbb{E}$, precomposition is an equivalence

$$\text{Fun}_{\text{ex,ps}}(\mathbb{L}, \mathbb{E}) \simeq \text{Fun}_{\text{ex,ps}, \mathcal{W}_B}(\mathbb{K}_!, \mathbb{E}). \quad (1)$$

Here functors are exact on Hom categories, transformations are pseudonatural (their naturality 2-cells are invertible), and modifications and their higher maps are retained. The right side is the full subcategory of functors taking every named refinement graph to an equivalence; it therefore inverts their saturation too. No Verdier quotient of an individual Hom category is meant. Theorem 9.5 constructs $\mathbb{L}$ as the full anchor category, rather than assuming a localization theorem. In this model the localized Hom category between the images of a, b is $\mathbb{K}_!(a_B, b_B)$, with its existing mapping spectra.

Derived Morita theory supplies the general relation between functors and bimodules, and the interpretation of Hochschild cohomology as endomorphisms of the identity functor; see Toën [5]. Integral-transform comparisons for quasicohherent sheaves on perfect derived stacks provide related geometric background [6]. That quasicohherent setting is different from our constructible nearby coefficient categories and does not supply their kernel comparison. The geometric contribution here is closure of the specified kernel family, invertibility of the

particular one-sided counit, and the compatibility of its actual comparison diagrams. Morita coordinates, opposite-bimodule duality and enhanced assembly are general mechanisms, not new general Morita or six-operation theorems.

The companion manuscripts in [1, 2] serve as fixed inputs for the dependencies listed below. The cited version dates identify the specific source versions used; these manuscripts are not bundled with this public release. Their original theorem labels are retained. The following are the precise geometric inputs from [2], in its fixed geometric rational coefficient domain:

1. **lem:branch-algebras** and **prop:unit-generation**: actual branch algebra maps and unipotent generation, with their module-valued evaluation maps; used in Proposition 5.3 and the geometric generators for Proposition 5.4;
2. **lem:cell-algebra-unit**: the actual equivalence $H_{S_I} \rightarrow Ru_{JI,*}H_{S_J}$ for complete inverse images of locally closed coarse supports, as an algebra map; used in the generated kernel convolution formulas and Theorem 8.3;
3. **thm:cell-reconstruction**: the intrinsic cartesian category on a complete coarse-stratum fibre is generated by its regular H -module; used in Proposition 5.3;
4. **lem:fibre-cartesian-ambient**: the actual adjunction $u_H^* \dashv Ru_*$ identifies the selected categories, with arbitrary coarse attachments and full module mapping spectra; used in Proposition 2.4 and the selected coefficient comparisons underlying Theorem 8.3;
5. the fixed integer-cut chart contractions and their actual compositions. Their algebra maps come from the one nearby functor.

No product-kernel descent or new convolution coherence is imported from that paper. External products, module bars and support base-change comparisons are constructed using the scheme six-operation enhancement of [3, 4]; this input is functorial on correspondence diagrams, including the pasting cells and adjunction mates, not just six triangulated functors. All cited local results are used only on the four fixed models and complete coarse supports.

Theorem 1.1 (Generated closure, one-sided descent and realization). *For the fixed data above and the source of Definition 2.5, the following hold.*

- (A) Geometric control. *The graph and rectangular families of Section 4 control the generated convolution source. Every generated kernel and every selected action integrand is bounded constructible, and its action preserves the specified coefficient type.*
- (B) One-sided descent. *For $a = (J_a, S_a)$, $b = (J_b, S_b)$, $c = (J_c, S_c)$, $I \subset J_a, J_b, J_c$, $P : a \rightarrow b$ and $L : b \rightarrow c$, product direct image T_I and the actual refinement graph counit give equivalences*

$$U_b \star! T_I(P) \xrightarrow{\sim} P \star! U_a, \quad T_I(L) \star! T_I(P) \xrightarrow{\sim} T_I(L \star! P).$$

Together with the actual unit they make T_I strongly compatible with composition. At $I = B$ they identify the refinement-localized generated source with its full generated source at the anchor.

- (C) Verdier-compatible Morita realization. *There is an enhanced realization into the independently constructed target of Definition 5.7,*

$$\mathbf{Sk}_{!,B}^{\ker,\min} : \mathbf{K}_{\log,k;B}^{\text{dec},\min,\Psi}[\mathcal{W}_B^{-1}] \longrightarrow \mathbf{K}_{\text{cell},k;B}^{\text{dec},\text{Mor},\text{rperf}}.$$

Writing $\mathbf{Sk}_!$ for this realization, its specified evaluation and Fubini maps give

$$\boxed{\mathbf{Sk}_!(L \star_! P) \simeq \mathbf{Sk}_!(L) \star \mathbf{Sk}_!(P)}, \quad \mathbf{Sk}_!(\Delta_* H_a) \simeq \mathbf{B}_S. \quad (2)$$

For $P : a \rightarrow b$, $L : b \rightarrow c$, the right side in Morita coordinates is $M_P \otimes_{\mathbf{B}_T} M_L$; this fixes the composition order. There is also a star realization on the corresponding supported source and an enhanced equivalence

$$D_{\text{cell}} \circ (\mathbf{Sk}_{!,B}^{\text{ker,min}})^{\text{co}} \simeq \mathbf{Sk}_{*,B}^{\text{ker,min}} \circ D_{\log,\Psi}.$$

The composition and unit constraints are coherently compatible with these actual Verdier mates, biduality, coarse support localization, and the support base-change and proper refinement diagrams used here, including the four-model diamond.

Here co reverses kernel morphisms, not their endpoints; $\mathcal{W}_B$ consists of the specified refinement graphs. No unrestricted base-change theorem or full faithfulness of this realization is asserted.

Part (A) is proved in Theorem 4.3. The central step of (B) is Theorem 8.3; Theorem 9.5 deduces localization from it. Part (C) uses the geometric mate and Morita comparison, assembled with descent in Proposition 9.4. The superscript Ψ is essential; the subscript $\log$ records the model's origin, not a proved original log/root kernel realization.

2 Geometric coefficient types and kernel sources

We use the enhanced constructible adic scheme formalism of [3, 4], on the scheme domains used in [2] and rationalized to $\mathbf{Q}_\ell$. For an ambient category in which arbitrary relative tensors can be formed, use $\mathcal{D}(S) = \text{Ind}(D_c^b(S, \mathbf{Q}_\ell))$, with the tensor, inverse-image and shriek direct-image functors extended continuously from constructible coefficients. Their right adjoints are taken in this ambient category. On the bounded constructible domains used below they agree with the actual geometric adjoints. Module mapping objects are mapping objects in this enhancement, not maps of underlying cohomology modules. Ind-completion is an ambient device; it does not assert constructibility of a relative tensor over a nearby algebra.

Lemma 2.1 (Bounded objects in the ambient completion). *For a constructible algebra A , the full subcategory of A -modules in $\mathcal{D}(S)$ with bounded constructible underlying object has the actual enhanced module structures and mapping spectra. The same holds for bimodules. Scalar geometric right adjoints, when their actual values are bounded constructible, agree with the right adjoints in this completion. Direct image of module sheaves has the same agreement on this domain.*

Proof. Write $\mathcal{C} = D_c^b(S, \mathbf{Q}_\ell)$. The embedding $\mathcal{C} \rightarrow \text{Ind}(\mathcal{C})$ is fully faithful and symmetric monoidal. The coherent data for an action on a bounded object involve finite tensor words in bounded objects; they therefore use the same mapping spectra in either enhancement. More explicitly, for bounded underlying M, N , the free-module bar resolution gives

$$\text{Map}_{\text{Mod}_A}(M, N) \simeq \text{Tot}_{[n] \in \Delta} \text{Map}_{\mathcal{C}}(A^{\otimes n} \otimes M, N).$$

This is the same cosimplicial spectrum for actual module sheaves and for the completion. Its totalization is taken in spectra; no commutation of a limit with a colimit is asserted. Applying the same argument to the tensor product of the two acting algebras, with the appropriate opposite algebra, proves the bimodule assertion.

For a scalar adjunction, test the ambient right adjoint against every $C \in \mathcal{C}$. The geometric adjunction identifies these maps with those into the actual bounded right-adjoint value. These C are compact generators of the completion, so they identify the two values. For a map with an algebra comparison $f^*A \rightarrow B$, test the right adjoint to module inverse image against free modules $A \otimes C$. Their inverse images are $B \otimes f^*C$; the free module adjunction and scalar geometric adjunction give the same representing value, with its restricted A -action, as actual module direct image. Free modules on the C generate the ambient module category, which proves the assertion. This does not claim that an arbitrary bounded-underlying module is compact as a module, or that an arbitrary relative tensor or relative internal Hom is bounded. Those boundedness statements must be established separately below. $\square$

Module conventions. Coefficient objects are right H_a -modules; supported objects are right H_a^{op} -modules, equivalently left H_a -modules. A shriek kernel is left H_a , right H_b . Dualizing its two commuting actions gives left H_b , right H_a ; we then exchange their written order, obtaining left H_a^{op} , right H_b^{op} . This exchange uses the symmetry of complexes with its Koszul signs. It changes algebra sides, not the geometric endpoints. For N supported, $p_a^!N$ is a left $p_a^*H_a$ -module: use $p_a^*H_a \otimes p_a^!N \rightarrow p_a^!(H_a \otimes N) \rightarrow p_a^!N$. Thus $R\mathcal{H}om_{p_a^*H_a}(P, p_a^!N)$ takes Hom of left modules. Precomposition with the right H_b -action on P gives a left H_b -action, written as a right H_b^{op} -action after pushforward. All exchanges below use this same convention on mapping spectra.

Definition 2.2 (Intrinsic coefficient category on one fibre). For a complete coarse-stratum fibre $Z \subset S_J$, let $\mathcal{C}_J(Z)$ be the full category of bounded constructible right H_Z -modules with unipotent lisse cohomology on each fine open stratum and perfect node restrictions. Each individual internal branch attaching map from a node module to its branch link must be an equivalence of node modules. This condition is imposed on each branch separately, not on their direct sum. Morphisms are the actual module mapping spectra. Exactness of restriction and links and closure of perfect/unipotent objects under retracts make this a stable idempotent-complete category. On U it is the unipotent lisse category; on a coarse node it is $\text{Perf}(A_q)$.

Definition 2.3 (Star and supported coefficient types). For a coarse support S , put $U_S = S \cap U$ and $Z_S = S \cap \{q_0, q_\infty\}$. The category $\mathcal{A}_B(S)$ is the full category of H_S -modules with bounded constructible underlying complex, unipotent lisse restriction to U_S , and restriction in $\text{Perf}(A_q)$ at each $q \in Z_S$, where $A_q = i_q^*H_S$. No invertibility is required of its attaching maps. On S_J , define $\mathcal{A}_{J|B}(S_J)$ by the same fine-stratum conditions, requiring branchwise cartesian coefficients only within the complete inverse image of each individual coarse stratum. Thus on the fibre $Z_{q,J}$ all its internal branch attachment maps are equivalences, whereas its attachments to the coarse open stratum are unrestricted. This is an intrinsic condition on actual module sheaves, not a condition of lying in a prescribed essential image.

Use ordinary Verdier duality

$$D_S M = R\mathcal{H}om_{\mathbf{Q}_\ell}(M, \omega_S), \quad \omega_S = a_S^! \mathbf{Q}_\ell.$$

The supported type $\mathcal{A}_{J|B}^D(S_J)$ is specified as follows. Its objects are bounded constructible opposite-action modules N whose exceptional restriction to each coarse-stratum fibre Z belongs to $D_Z \mathcal{C}_J(Z)$. On the coarse open stratum this means the ordinary Verdier partner of the unipotent lisse category. Here $\mathcal{C}_J(Z)$ is specified by the intrinsic internal branch conditions just stated. Empty supports have the zero category.

Proposition 2.4 (The two coefficient types are Verdier partners). *Ordinary geometric Verdier duality induces an exact anti-equivalence*

$$D_{S_J} : \mathcal{A}_{J|B}(S_J)^{op} \simeq \mathcal{A}_{J|B}^D(S_J),$$

with the actual evaluation biduality. In particular no stalk-perfect condition is substituted for an exceptional-restriction condition on the supported side.

Proof. For the locally closed inclusion $e : Z \rightarrow S_J$, geometric duality gives $e^! D_{S_J} M \simeq D_Z e^* M$. This identifies exactly the specified conditions on each coarse-stratum fibre. Conversely $e^* D_{S_J} N \simeq D_Z e^! N$ identifies the reverse conditions. Verdier biduality on bounded constructible $\mathbf{Q}_\ell$ -complexes gives the inverse equivalences. Applying duality to the module action produces the opposite action; the biduality map is linear because evaluation is balanced for that action. The argument uses geometric duality, not tensor-dualizability as an H -module or as a sheaf. $\square$

The convention of [2] is $\tilde{D}_S = D_S(-1)[-2]$. Thus $D_S = \tilde{D}_S(1)[2]$. These categories are stable under Tate twists and shifts, but this translation must still be made on named support objects and pairings: $W_S := D_S H_S(-1)[-2]$, so $D_S H_S = W_S(1)[2]$. Globally $R\Gamma(S, D_S M) \simeq D_{\mathbf{Q}_\ell} R\Gamma_c(S, M)$. The regular unit is H_S , not $D_S H_S$ or W_S .

Definition 2.5 (Actual kernel categories and fixed generators). An object is a decorated support $a = (J, S)$, with coefficient type $\mathcal{A}_a = \mathcal{A}_{J|B}(S_J)$ and algebra H_a . For a, b , put

$$\mathcal{K}_1(a, b) = {}_{p_a^* H_a} \text{Bimod}_{p_b^* H_b}(\mathcal{D}(S_J \times_k T_K)).$$

Let $\mathcal{K}_1^b(a, b)$ be its full subcategory on kernels whose underlying $\mathbf{Q}_\ell$ -complex is bounded constructible. It is not restricted by a condition on the kernel's action. All kernel mapping spectra, including noninvertible morphisms, are retained.

Specify the following geometric generators.

1. The unit $\Delta_* H_a$ at every decorated support.
2. For a coarse open or closed inclusion $f : a \hookrightarrow b$ within one model, both graph kernels $\Gamma_{f,!} H_a$ on $a \times b$ and $\Gamma_{f,!}^t H_a$ on $b \times a$. Their actions are respectively $f_!$ and the module inverse image. The coefficient maps are the actual restriction isomorphisms. Also specify the diagonal images of $j_! H_U \rightarrow H_b \rightarrow i_* H_Z \rightarrow j_! H_U[1]$ for each coarse open/closed decomposition, with these actual arrows.
3. For each J and each endpoint q , the two rectangular kernels $H_{U_J} \boxtimes H_{Z_{q,J}}$ and $H_{Z_{q,J}} \boxtimes H_{U_J}$.
4. For $u : a' \rightarrow a$ a refinement on a complete inverse-image support, the transposed graph kernel $\Gamma_{u,*}^t H_{a'}$ on $a \times a'$, with the left action through $u^* H_a \rightarrow H_{a'}$. These are the distinguished arrows $\mathcal{W}_B$. Their actions are the actual u_H^* .

The proper graph in the reverse direction is available in the ambient kernel category, but is not declared an inverse kernel or added to this generating list. Proper direct image is already the inverse on the selected coefficient categories; inverting a refinement in the eventual source is a separate localization step.

The generated shriek source $\mathcal{K}_1^{\text{gen}}$ is the smallest full stable idempotent-complete family of subcategories of $\mathcal{K}_1$, containing this list and closed under Tate twists and finite composable convolutions. Equivalently take finite expressions in these operations, evaluate them in the ambient categories, and retain all actual maps between their evaluations. No quotient by equality of actions and no preservation test enters this definition.

We denote this enhanced generated source, with its fixed geometric decorations, by $\mathbf{K}_{\log,k;B}^{\text{dec,min},\Psi}$. The supported source uses the actual Verdier labels specified below; its product-duality functor is denoted $D_{\log,\Psi}$.

Proposition 2.6 (Ambient composition and geometric generator actions). *Ambient convolution and $\Delta_* H_a$ define associative unital composition on these kernel categories with their enhanced cells. The graph actions in Definition 2.5 are the named geometric functors. These assertions do not imply that $\mathcal{K}_1^{\text{gen}} \subset \mathcal{K}_1^b$, or that all its actions preserve $\mathcal{A}_a$.*

Proof. Form the relative tensor by its two-sided bar realization. Inverse image, tensor and shriek direct image preserve colimits in the specified ambient model. The ordinary geometric projection and base-change maps therefore extend to this realization. On a fourfold product, either iterated convolution is the direct image of the same three factors, relatively tensored over the two middle algebras. Fubini and tensor associativity give its comparison. For any number of factors use the same multiple bar realization and the composite projection; its substitution cells give the associativity coherences. The diagonal projection formula and the regular-module unit give both unit cells and their compatibilities. This is the ambient enhanced construction, not a boundedness argument for its relative tensors.

For the graph of f , restriction of the two projections is $(1, f)$; the projection formula cancels the regular source algebra and the remaining direct image is $f_!$. For the transposed graph, the projections restrict to $(f, 1)$, giving $H_a \otimes_{f^* H_b} f^*(-)$. These calculations use the same units and counits as the geometric functors, hence identify their action on morphisms too. Each listed generator has bounded constructible underlying complex: its coefficient is an actual constructible nearby restriction, and the specified maps are finite-type immersions or proper graphs. This verifies the generators, not their convolution closure. $\square$

Definition 2.7 (Bounded star labels and their variance). The supported object corresponding to a is a^D with coefficient category $\mathcal{A}_a^D$. The bounded star kernel category from a^D to b^D consists of geometric labels $P^D = D_{S_J \times T_K} P$ for $P \in \mathcal{K}_1^b(a, b)$, with their actual opposite-action module maps. This is a full subcategory of left $p_a^* H_a^{\text{op}}$, right $p_b^* H_b^{\text{op}}$ modules. The generating star labels are the duals of the specified generators. Biduality gives

$$(\mathcal{K}_1^b(a, b))^{\text{op}} \simeq \mathcal{K}_*^b(a^D, b^D).$$

It changes the two coefficient types, not the order of the endpoints. If a shriek composite is bounded constructible, its star-labelled composite is $D(L \star_! P)$. This specifies the star composition convention on its bounded domain. Boundedness of every generated composite is proved separately in Theorem 4.3.

Indeed duality reverses the two commuting actions; interpreting them as actions of the opposite algebras in the displayed order gives the stated types. Geometric biduality identifies the full module mapping spectra and their reversed composition. The star-labelled unit is $D\Delta_* H_a$, not an assertion that DH_a is the shriek unit.

The actual star action to verify is

$$N \longmapsto R p_{b,*} R \mathcal{H}om_{p_a^* H_a}(P, p_a^! N), \quad N \in \mathcal{A}_a^D, \quad P = D(P^D).$$

This expression is covariant in the label P^D , since its Hom is contravariant in P . It is defined in the ambient module categories. Its selected bounded domain and geometric identification are proved below, rather than inserted in the definition of an allowed kernel. Theorem 4.3 proves that the generated source is bounded, and Theorem 6.1 identifies this actual geometric operation on its selected coefficients.

3 Geometric kernels and a first convolution calculation

For coarse supports S, T write p_S, p_T for the projections of $S \times_k T$. A kernel has commuting left $p_S^* H_S$ and right $p_T^* H_T$ actions; input modules are right modules. Its shriek action is the

actual functor

$$\Phi_P^!(M) = Rp_{T,!}(p_S^* M \otimes_{p_S^* H_S}^L P).$$

For kernels $P : S \rightarrow T$ and $L : T \rightarrow V$, the actual ambient convolution is

$$L \star! P = Rp_{SV,!}(p_{ST}^* P \otimes_{p_T^* H_T}^L p_{TV}^* L).$$

These expressions are initially formed in the ambient derived module categories. Bounded constructibility and preservation of the selected coefficient categories require proofs for the prescribed generators.

Definition 2.5 fixes the generators and their ambient closure. In this section we start checking their actual integral transforms; later preservation arguments must apply to that specified collection rather than redefine it by the desired conclusion.

Proposition 3.1 (Ambient rectangular convolution). *At the coarse model, put $P_{S,T} = H_S \boxtimes_{\mathbf{Q}_\ell} H_T$ for $S, T \in \{U, q\}$, where q is either endpoint. The geometric projection and compact-support base-change maps give*

$$\Phi_{P_{S,T}}^!(M) \simeq R\Gamma_c(S, M) \otimes_{\mathbf{Q}_\ell} H_T$$

for bounded constructible module coefficients M , and

$$P_{T,V} \star! P_{S,T} \simeq R\Gamma_c(T, H_T) \otimes_{\mathbf{Q}_\ell} P_{S,V}.$$

These are natural ambient comparisons, with the indicated outer coefficient actions.

Proof. In the action formula the first coefficient factor of the rectangle is the free regular H_S -module. The relative tensor is consequently $M \boxtimes H_T$. Compact-support base change for $S \times T \rightarrow T$, followed by the projection formula, gives the first map and identifies it as an equivalence. On $S \times T \times V$, the convolution integrand is canonically

$$p_S^* H_S \otimes_{\mathbf{Q}_\ell} p_T^* H_T \otimes_{\mathbf{Q}_\ell} p_V^* H_V,$$

because the two middle regular factors tensor over $p_T^* H_T$. Compact-support base change and projection along the middle factor give the second comparison. These are maps in the same enhanced six-operation formalism, hence natural in kernel and coefficient maps. On these coarse strata $H_U = \mathbf{Q}_{\ell,U}$ by smooth specialization and H_q is the bounded finite-dimensional node cochain algebra. Thus the displayed compact-support complexes are bounded and finite. The tensor calculation cancels a regular algebra factor, not a Verdier dual of an attached coefficient. $\square$

Write $A_q = H_q$ as its actual node cochain algebra. The two tests give

$$P_{q,U} \star! P_{U,q} \simeq A_q \otimes_{\mathbf{Q}_\ell} P_{U,U}, \quad P_{U,q} \star! P_{q,U} \simeq R\Gamma_c(\mathbf{G}_m, \mathbf{Q}_\ell) \otimes_{\mathbf{Q}_\ell} P_{q,q}.$$

Here $H^0(A_q) = \mathbf{Q}_\ell$, $H^1(A_q) = \mathbf{Q}_\ell(-1)$, while $H_c^1(\mathbf{G}_m, \mathbf{Q}_\ell) = \mathbf{Q}_\ell$ and $H_c^2(\mathbf{G}_m, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-1)$; other degrees vanish. We retain the actual complexes and maps, without choosing a splitting or replacing them by Euler characteristics. Its compatibility with the localization boundary is proved in Proposition 6.2; actual fine-model product descent is proved in Section 8.

4 A finite geometric family controlling all generated kernels

For $W \subset S$ a locally closed union of coarse strata, write $e_{W,S}^J : W_J \rightarrow S_J$ and put

$$E_S^J(W) = e_{W,S,!}^J H_{W_J}.$$

These are actual coefficient modules, with their restriction-induced H_{S_J} -actions. Extension by zero for a locally closed immersion uses its usual open/closed factorization. All intersections below are scheme-theoretic complete inverse images of coarse intersections. In particular their maps are not arbitrary fine support inclusions.

For $J \subset K$, supports S, T , and $W \subset S \cap T$, let

$$\gamma_W : W_K \longrightarrow S_J \times T_K, \quad z \longmapsto (e_{W,S}^J u_{KJ}(z), e_{W,T}^K(z)), \quad G_{J,K}^{S,T}(W) = \gamma_{W,!} H_{W_K}.$$

The source action uses the actual nearby algebra pullback map; the target action is restriction. Also set, for $W \subset S$, $Z \subset T$,

$$R_{J,K}^{S,T}(W, Z) = E_S^J(W) \boxtimes_{\mathbf{Q}_\ell} E_T^K(Z).$$

Call these the graph and rectangular families. They are defined geometrically on products, independently of the actions they induce. Empty supports give zero. In formulas we suppress the outer support superscripts when the endpoints specify them.

Lemma 4.1 (Derived support intersection). *Let H be the nearby algebra on a support X , and $e : Z \rightarrow X$, $f : V \rightarrow X$ allowed locally closed immersions. For $h : Z \cap V \rightarrow X$ there is a canonical bimodule equivalence*

$$(e_! e^* H) \otimes_H^L (f_! f^* H) \simeq h_! h^* H.$$

It includes empty, open, closed and mixed supports, and is natural in their support base-change diagrams.

Proof. Set $Q_Z = e_! \mathbf{Q}_{\ell,Z}$ and $Q_V = f_! \mathbf{Q}_{\ell,V}$, regarding these as scalar sheaves on X . Scalar projection for immersions identifies $e_! e^* H = H \otimes_{\mathbf{Q}_\ell} Q_Z$ as an H -bimodule. The relative bar therefore cancels a free regular factor and gives $H \otimes_{\mathbf{Q}_\ell} Q_Z \otimes_{\mathbf{Q}_\ell} Q_V$. The scalar map $Q_Z \otimes Q_V \rightarrow h_! \mathbf{Q}_{\ell,Z \cap V}$ is the support base-change/projection map; on every geometric stalk it is $\mathbf{Q}_\ell \otimes \mathbf{Q}_\ell \rightarrow \mathbf{Q}_\ell$ or $0 \rightarrow 0$, hence an equivalence. Scalar coefficients form a field, so no hidden scalar Tor term is present. This proves the assertion with both regular actions. Projection and base change for the open/closed factorizations give the displayed map and its naturality. The bar proof works in the ambient completion by colimit extension of these scalar maps. In particular this is a sheaf support calculation, not a claim about derived intersections of coherent structure sheaves. $\square$

Proposition 4.2 (Four geometric convolution formulas). *For $J \subset K \subset L$, all well-typed instances of the following are canonical equivalences of kernels with both coefficient actions:*

$$\begin{aligned} G_{K,L}(V) \star! G_{J,K}(W) &\simeq G_{J,L}(W \cap V), \\ R_{K,L}(V, Y) \star! G_{J,K}(W) &\simeq R_{J,L}(W \cap V, Y), \\ G_{K,L}(V) \star! R_{J,K}(W, Z) &\simeq R_{J,L}(W, Z \cap V), \\ R_{K,L}(V, Y) \star! R_{J,K}(W, Z) &\simeq C_K(Z \cap V) \otimes_{\mathbf{Q}_\ell} R_{J,L}(W, Y), \quad C_K(A) = R\Gamma_c(A_K, H_{A_K}). \end{aligned}$$

The scalar complex $C_K(A)$ is perfect over $\mathbf{Q}_\ell$. These are the base-change, projection and algebra-unit comparisons themselves, not identifications chosen from equality of the corresponding actions.

Proof. First, on one support the regular coefficient factors give

$$E_T^K(Z) \otimes_{H_{T_K}}^L E_T^K(V) \simeq E_T^K(Z \cap V).$$

This is Lemma 4.1, applied with both regular coefficient actions. It does not use perfectness over a product algebra. The same identity holds after pullback to a product of supports. Across a refinement we use only the actual algebra map $u^*H_{W_K} \rightarrow H_{W_L}$, not an assertion that it is an isomorphism. The regular cancellation there is $H_{W_L} \otimes_{u^*H_{W_K}} u^*H_{W_K} = H_{W_L}$. Each support base-change map below is the scalar shriek Beck–Chevalley map, extended to the relative bar. The source and target actions commute with every face and degeneracy of that bar; hence these comparisons, their inverses, and their Fubini pasting cells are bimodule maps.

For the first formula the two graph supports on the threefold product have fibre product $(W \cap V)_L$. On that fibre product the middle regular algebra is pulled back from level K , and its tensor with $H_{(W \cap V)_L}$ cancels by the regular-module unit. Composition of the two monomial model maps gives exactly the stated graph. Shriek base change on the graph squares and composition of direct images construct the asserted kernel map.

For the second formula, after the same cancellation and intersection, the remaining source factor is the direct image of $H_{(W \cap V)_K}$ along $(W \cap V)_K \rightarrow (W \cap V)_J$ and then its immersion into S_J . This map between complete inverse-image supports is proper. The actual nearby algebra unit of [2] gives

$$H_{(W \cap V)_J} \xrightarrow{\sim} Ru_*H_{(W \cap V)_K}.$$

Its inverse, with the specified algebra actions, supplies the required comparison. Here is its linearity as a specified map. Put $A = H_{(W \cap V)_J}$, $D = Ru_*H_{(W \cap V)_K}$ and let $\eta : A \rightarrow D$ be that algebra unit. Regard D as an A -bimodule through η , and let $r = \eta^{-1}$ in that enhanced bimodule category. Multiplicativity of η and its inverse give the square

$$\begin{array}{ccc} A \otimes D & \xrightarrow{1 \otimes r} & A \otimes A \\ m_D \downarrow & & \downarrow m_A \\ D & \xrightarrow{r} & A. \end{array} \quad (3)$$

Here m_D is the action through η ; the right-action square is obtained in the same way. These are squares in the enhancement: an equivalence of algebra objects has its inverse there, rather than an arbitrarily chosen inverse on cohomology. Restriction of the source algebra and extension by zero retain these squares. External product with $E^L(Y)$ retains target linearity, since $1 \boxtimes m_Y$ commutes with $r \boxtimes 1$ by tensor functoriality. Thus both outer actions on the second formula are accounted for. External product with $E^L(Y)$ then gives the second formula. This is where a genuine proper coefficient comparison is used; an ordinary graph pullback assertion would not suffice.

For the third formula the middle variable is fixed by the graph at level L . Its coefficient restriction is the module pullback of $E^K(Z)$, intersected with V_L , hence $E^L(Z \cap V)$. The source factor $E^J(W)$ remains unchanged. For the fourth formula the middle tensor is the first displayed intersection module. Compact-support base change and projection integrate it to $C_K(Z \cap V)$. All steps are maps of the actual sheaf modules on the threefold product and respect the outer actions.

The coefficient H_{A_K} is bounded constructible, the support is separated of finite type over the algebraically closed field, and coefficients are geometric rational ℓ -adic coefficients. Constructible compact-support finiteness in the specified six-operation domain therefore makes $C_K(A)$ a bounded finite-dimensional complex. Alternatively proper pushforward to A_B identifies it with $R\Gamma_c(A_B, H_{A_B})$. No compact-support complex is replaced by its Euler characteristic. $\square$

Theorem 4.3 (Bounded kernels and preservation of the selected types). *Every kernel in $\mathcal{K}_!^{\text{gen}}(a, b)$ has bounded constructible underlying complex. Its actual integral transform is an exact functor $\mathcal{A}_a \rightarrow \mathcal{A}_b$, natural in all actual kernel maps. Moreover, for $M \in \mathcal{A}_a$ the integrand $p_a^* M \otimes_{p_a^* H_a}^L P$ is bounded constructible on the product. These assertions apply to the fixed generating collection and its whole finite convolution closure; they are not admissibility conditions.*

Proof. For each comparable pair $J \subset K$, let $\mathcal{N}(a, b)$ be the full thick subcategory of the ambient kernel category generated by all the graph and rectangular families above, allowing perfect $\mathbf{Q}_\ell$ -scalar factors. For a noncomparable pair use the zero category. Every specified generator belongs to this family: the unit and the localization diagonal terms are $G_{J,J}(W)$, the immersion graphs are its corresponding support-to-support versions, refinement graphs are $G_{J,K}(S)$, and the given rectangles are instances of R . Each family member is bounded constructible. The four formulas and perfectness of $C_K(A)$ show that $\mathcal{N}$ is closed under convolution on its generators. Since ambient convolution is exact in each variable and preserves retracts, this extends first in one variable and then the other to all of $\mathcal{N}$. Thus the independently defined generated source is contained in $\mathcal{N}$, and every one of its objects is bounded constructible. There is no assertion of a uniform amplitude for all finite words.

For preservation compute the family actions before taking any thick closure. The graph action is

$$\Phi_{G_{J,K}(W)}^!(M) \simeq e_{W,T,!}^K(u_{KJ,W,H}^* e_{W,S}^{J,*} M),$$

and the rectangular action is

$$\Phi_{R_{J,K}(W,Z)}^!(M) \simeq R\Gamma_c(W_J, e_{W,S}^{J,*} M) \otimes_{\mathbf{Q}_\ell} E_T^K(Z).$$

The graph calculation is projection along its actual graph; the rectangular one is compact-support base change and cancellation of the regular source factor. Restriction to a coarse support union preserves the intrinsic fibre conditions. Actual module pullback under refinement preserves them by the ambient-gluing theorem of [2]. Extension by zero across coarse strata preserves them too: a whole coarse fibre is either retained or is assigned zero, so no internal fibre attachment is cut apart. Hence the graph action lies in $\mathcal{A}_b$. For the rectangle the scalar factor is perfect over $\mathbf{Q}_\ell$, while $E_T^K(Z)$ belongs to $\mathcal{A}_b$ by these same restriction and extension observations. This proves its preservation. Exactness and retracts extend the assertion to $\mathcal{N}$ because the selected coefficient categories are stable and idempotent-complete.

For completeness the stronger integrand assertion is checked before pushing forward. For a graph it is the graph extension of $u_{KJ,W,H}^* e_{W,S}^{J,*} M$, which has just been shown to be a bounded selected coefficient. For a rectangle it is

$$(e_{W,S,!}^J e_{W,S}^{J,*} M) \boxtimes_{\mathbf{Q}_\ell} E_T^K(Z),$$

a bounded constructible external product. Exactness of the integrand functor in the kernel extends this to $\mathcal{N}$. This explicitly rules out an unbounded relative Tor integrand in the claimed domain. All functors and comparison maps were defined in the ambient module categories, so preservation restricts them on their full mapping spectra; it does not select a quotient of kernel morphisms. $\square$

Remark 4.4 (Why the direction of refinement matters). The proof uses the generating direction of Definition 2.5: refinement acts by u_H^* . It does not admit an arbitrary reverse graph before localization. Over a contracted fibre the latter would involve the coefficient map $A_q \rightarrow H_z$ in the other module direction; tensor-perfectness over A_q does not follow from finite-dimensionality over $\mathbf{Q}_\ell$. The four-formula argument, rather than an unrestricted finite-Tor claim, is the closure proof used here.

5 Cellular coefficients from strata and links

We now construct the cellular category without defining it as a geometric endomorphism algebra. All data in the construction are stratum and punctured-branch data at the coarse model. Put $A = R\Gamma(U, \mathbf{Q}_\ell)$ and, for each endpoint q , let B_q be the cochain algebra of the punctured strict neighbourhood on its included branch. Retain the actual algebra maps

$$A \xrightarrow{\kappa_q} B_q \xleftarrow{\lambda_q} A_q.$$

These are equivalences by the branch-algebra calculation in [2], with the primitive Kummer class and the sign of the chosen branch coordinate. They are not replaced by chosen identifications of their graded cohomology rings. Regard B_q as an (A, A_q) -bimodule through these maps and define

$$\mathcal{L}_q(N) = N \otimes_A^L B_q \quad (N \in \text{Perf}(A)).$$

It takes values in $\text{Perf}(A_q)$ because both algebra maps are equivalences. This is a branch functor, not a condition on attachments.

Definition 5.1 (The cellular gluing category). If $U \subset S$, define $\mathcal{E}_B(S)$ to be the stable homotopy-coherent comma category of data

$$(N, (M_q), (\alpha_q)), \quad N \in \text{Perf}(A), \quad M_q \in \text{Perf}(A_q), \quad \alpha_q : M_q \longrightarrow \mathcal{L}_q(N) \quad (q \in Z_S).$$

Precisely, it is the pullback of the arrow category of $\prod_{q \in Z_S} \text{Perf}(A_q)$ along the functor sending $(N, (M_q))$ to $((M_q), (\mathcal{L}_q(N)))$. If S has only endpoints, set $\mathcal{E}_B(S) = \prod_{q \in S} \text{Perf}(A_q)$; for $S = \emptyset$ use the zero category. These constructions are stable and idempotent-complete: the arrow category and the displayed exact pullback compute finite limits, colimits and retracts componentwise.

For two triples E, E' , the full mapping spectrum is

$$\begin{aligned} \text{Map}_{\mathcal{E}}(E, E') &= \text{fib} \left(\begin{array}{c} \text{Map}_A(N, N') \oplus \prod_q \text{Map}_{A_q}(M_q, M'_q) \\ \longrightarrow \prod_q \text{Map}_{A_q}(M_q, \mathcal{L}_q N') \end{array} \right), \\ (f, (g_q)) &\longmapsto (\mathcal{L}_q(f)\alpha_q - \alpha'_q g_q)_q. \end{aligned} \tag{4}$$

This formula includes the homotopies in the attaching squares. Deleting an endpoint deletes its data. Restriction to the open stratum selects N ; restriction to an endpoint selects M_q . The closed costalk in the reconstructed geometry is $\text{fib}(\alpha_q)$, not M_q .

Lemma 5.2 (Module recollement and the selected generators). *The open/closed gluing and mapping formulas used here hold in actual right H -module sheaves. On U , bounded complexes with unipotent lisse cohomology form $\text{thick}(\mathbf{Q}_{\ell, U})$. On a complete coarse-node fibre $v : Z \rightarrow q$, the functor $F_q = H_Z \otimes_{A_q} (-)$ identifies $\text{Perf}(A_q)$ with $\mathcal{C}_J(Z)$.*

Proof. The actions on Rj_*V and i_*M use respectively the algebra units $H \rightarrow Rj_*j^*H$ and $H \rightarrow i_*i^*H$. Naturality of these units makes $Rj_*V \rightarrow i_*i^*Rj_*V$ linear. An attaching map $M \rightarrow i^*Rj_*V$ is a map of i^*H -modules, so its direct image is linear too. Forgetting module actions creates limits: an action on a limit is its compatible family of actions, including their associativity coherences. Thus the finite homotopy pullback of these two linear maps is the sheaf pullback with that action. The pair (j^*, i^*) is conservative on modules, because it is so on underlying sheaves and the forgetful functor detects equivalences. This proves the gluing equivalence before taking Morita coordinates.

Mapping spectra in this pullback description are the homotopy fibre of the difference of the two attaching compositions, giving (4) after the branch identifications. The linear triangle $j_!V \rightarrow Rj_*V \rightarrow i_*i^*Rj_*V$ pulled back along $i_*M \rightarrow i_*i^*Rj_*V$ gives the localization triangle and its actual connecting arrow. These constructions use finite limits and the actual bounded values of Rj_* and i_* ; Lemma 2.1 identifies them in the completion. No commutation of an unbounded relative bar with Rj_* or $i^!$ is asserted. The left functors and their module projection/base change are obtained from scalar projection on free modules and bar realization; their bounded values here were separately checked.

A unipotent local system has a finite filtration with constant successive quotients. Each lies in $\text{thick}(\mathbf{Q}_{\ell,U})$; finite cohomological truncations give the assertion for bounded complexes. Conversely long exact sequences and retracts preserve unipotent lisse cohomology. This also specifies the enhanced open category as a full subcategory, not only an abelian filtration class.

Finally the actual algebra unit $A_q \rightarrow Rv_*H_Z$ is invertible. The adjunction unit on a free A_q -module is this map; projection, finite cones and retracts prove it for every perfect module. Hence F_q is fully faithful there. Its essential image is $\text{thick}(H_Z)$, which is the intrinsic fibre category by `thm:cell-reconstruction` of [2]. This proves essential surjectivity and the selected counit, rather than inferring an equivalence of module categories from an algebra comparison alone. $\square$

Proposition 5.3 (Enhanced geometric reconstruction). *There is a natural equivalence*

$$\mathcal{A}_B(S) \simeq \mathcal{E}_B(S).$$

It identifies full module mapping spectra, the actual open/closed localization triangle and its connecting morphism. The attachment α_q is unrestricted.

Proof. Lemma 5.2 identifies the selected sheaves on U with $\text{Perf}(A)$ via $R\Gamma(U, -)$: the unit generates, and the mapping comparison on it is its actual cochain algebra. Let V_N denote the inverse Morita realization. The branch calculation gives the actual evaluation equivalence

$$R\Gamma(U, V_N) \otimes_A B_q \xrightarrow{\sim} i_q^*Rj_*V_N$$

as A_q -modules. It is the restriction/evaluation map on the unit; exactness and retracts prove it on all perfect N . Thus it identifies the independently specified $\mathcal{L}_q$ with the geometric link, including its action.

Given a triple, form in actual H_S -modules

$$F = Rj_*V_N \bigotimes_{i_*L(V_N)}^h i_*M, \quad L(V_N) = \prod_{q \in Z_S} i_q^*Rj_*V_N.$$

The first H_S -action is induced by the open-adjunction algebra map, the other two by its closed restrictions. Forgetting module actions creates this finite limit. Its restrictions are V_N and M_q . The open coefficient and all its links are bounded constructible; perfect A_q -modules are bounded finite-dimensional over $\mathbf{Q}_{\ell}$. Consequently Rj_*V_N and the finite pullback are bounded constructible, and F satisfies exactly Definition 2.3. No condition of invertibility of α_q was used.

Conversely an actual module has the attachment obtained from $F \rightarrow Rj_*j^*F$. The natural map to the displayed pullback is an equivalence after j^* and i^* , hence an equivalence by localization. Recollement in modules gives exactly (4), with A_q -linear maps in the last term. Thus these inverse constructions are enhanced equivalences, rather than bijections on

isomorphism classes. The triangle $j_!V_N \rightarrow F \rightarrow i_*M$ is the geometric localization triangle; its connecting morphism corresponds to α under

$$\mathrm{Map}_{H_S}(i_*M, j_!V_N[1]) \simeq \mathrm{Map}_{A_{Z_S}}(M, L(V_N)).$$

Here the identification is induced by the connecting arrow $i_*L(V_N) \rightarrow j_!V_N[1]$ of $j_!V_N \rightarrow Rj_*V_N \rightarrow i_*L(V_N)$. Pulling this triangle back along $i_*\alpha$ gives the triangle for F , so its boundary is that connecting arrow composed with $i_*\alpha$. This fixes the sign and rotation convention without choosing an extension class from an abstract isomorphism of groups. Endpoint-only and empty supports give the asserted product and zero categories directly. $\square$

Proposition 5.4 (Generators and the cellular Morita algebra). *Let $g_U = (A, 0, 0)$ and $g_q = (0, A_q, 0)$ in the cellular category, including just the strata present in S , and let g_S be their finite direct sum. Then $\mathcal{E}_B(S) = \mathrm{thick}(g_S)$. Its independently defined algebra $B_S = R\mathrm{Hom}_{\mathcal{E}}(g_S, g_S)$ satisfies*

$$\mathcal{E}_B(S) \simeq \mathrm{Perf}(B_S), \quad B_S \simeq R\mathrm{Hom}_{H_S}(G_S, G_S), \quad G_S = j_!\mathbf{Q}_{\ell, U} \oplus \bigoplus_{q \in Z_S} i_{q,*}A_q.$$

Omit absent summands. If $U \subset S$, the algebra has blocks

$$B_S \simeq \begin{pmatrix} A & \bigoplus_{q \in Z_S} B_q[-1] \\ 0 & \prod_{q \in Z_S} A_q \end{pmatrix},$$

where rows index targets and columns index sources. This denotes the derived triangular algebra with the actual branch bimodule actions and their composition, not its cohomology algebra.

Proof. Each triple has the componentwise cofiber sequence $(N, 0, 0) \rightarrow (N, M, \alpha) \rightarrow (0, M, 0)$. The first and last terms are generated by g_U and the g_q respectively, since their coefficients are perfect modules over the corresponding vertex algebras. Hence g_S generates; conversely these generators and their thick closure satisfy the defining conditions. Formula (4) gives

$$\begin{aligned} R\mathrm{Hom}(g_U, g_U) &= A, & R\mathrm{Hom}(g_U, g_q) &= 0, \\ R\mathrm{Hom}(g_q, g_U) &= B_q[-1], \\ R\mathrm{Hom}(g_q, g_{q'}) &= \begin{cases} A_q & q = q', \\ 0 & q \neq q'. \end{cases} \end{aligned}$$

The off-diagonal term is the fibre of $0 \rightarrow B_q$. Its left A and right A_q actions are the branch bimodule actions; there are no two composable off-diagonal blocks. This constructs the displayed triangular algebra from the cellular data. The enhanced reconstruction sends g_U to $j_!\mathbf{Q}_{\ell, U}$ and g_q to $i_{q,*}A_q$, so it identifies algebras with their multiplication. Finally the Yoneda comparison for $R\mathrm{Hom}(g_S, -)$ is an equivalence on g_S . Exactness and retracts extend it in each variable to its thick closure, proving full faithfulness. Its image is the thick closure of the regular module, namely $\mathrm{Perf}(B_S)$. The same proof works for endpoint-only supports. $\square$

In particular the endpoint generator is $i_{q,*}A_q$, not $i_{q,*}\mathbf{Q}_{\ell}$, and the total algebra is not $R\Gamma(S, H_S)$. The latter describes the smaller unit-generated cartesian category; it misses the independent endpoint summands and their attachments.

Proposition 5.5 (Fine-model coefficient generators). *For $u = u_{JB} : S_J \rightarrow S$, actual module pullback $F_J = H_{S_J} \otimes_{u^* H_S} u^*(-)$ identifies $\mathcal{E}_B(S)$ with $\mathcal{A}_{J|B}(S_J)$. Its generator is*

$$G_{J,S} = F_J G_S \simeq j_{J!} \mathbf{Q}_{\ell, U_J} \oplus \bigoplus_{q \in Z_S} i_{q,J,*} H_{Z_{q,J}},$$

and its endomorphism algebra is canonically $\mathbf{B}_S$. These coefficient comparisons respect the actual attachment boundary and the four-model refinement diagram. They make no statement about kernel categories on products.

Proof. Use the ambient-gluing comparison of [2], Lemma `lem:fibre-cartesian-ambient`. Its hypotheses hold: the models are integer cuts, the contraction is proper, the supports are full inverse images, and the coefficients on each contracted fibre satisfy the internal branchwise cartesian conditions. Over U it is the identity. Over q its left adjoint is $F_q(M) = H_{Z_{q,J}} \otimes_{A_q} M$, and the actual algebra unit $A_q \simeq Ru_{q,*} H_{Z_{q,J}}$ makes this an equivalence onto the intrinsic fibre category by its geometric unit generation. An arbitrary coarse attachment is sent to the actual mate $F_q L(V) \rightarrow L'(V)$ after $F_q \alpha$; the comparison uses $Ru_{q,*} L'(V) \simeq L(V)$ and does not assume $L'(V)$ itself is cartesian on the fibre. This is precisely the full mapping and boundary comparison proved there. The same base-change formulas give the displayed images of $j_!$ and i_* , hence the generator. Full faithfulness identifies its endomorphism algebra by the actual functor. The nearby algebra maps, pullback associativity and the composite-unit identity give the coefficient refinement cells, as in that paper's anchored theorem. None of these steps takes a product of kernel supports. $\square$

5.1 Dual cellular coefficients and the Morita target

Construct $\mathcal{E}_B^D(S) = \mathcal{E}_B(S)^{op}$ directly from the cellular category. Equivalently it is the opposite comma diagram: the vertex categories are $\text{Perf}(A)^{op}$ and $\text{Perf}(A_q)^{op}$, and each attachment arrow is reversed with all its homotopies. This construction precedes geometric realization and retains the entire link bimodule, not just duals of vertex cohomology groups. The label g_S^D is the object corresponding to g_S in this opposite category. The opposite of a stable category is stable, with cofiber triangles reversed and rotated in the stable convention.

Proposition 5.6 (Dual generators and their actual realization). *There are equivalences*

$$\mathcal{E}_B^D(S) \simeq \mathcal{A}_{J|B}^D(S_J) \simeq \text{Perf}(\mathbf{B}_S^{op}), \quad R\text{Hom}(D_{S_J} G_{J,S}, D_{S_J} G_{J,S}) \simeq \mathbf{B}_S^{op}.$$

They are induced by actual Verdier duality and carry the cellular biduality to geometric biduality. The dual generator is $G_{J,S}^D = D_{S_J} G_{J,S}$, not an assumed coefficient dual.

Proof. The opposite cellular category is generated by g_S^D , since opposites exchange finite fibers and cofibers and preserve retracts. Its endomorphism algebra is $\mathbf{B}_S^{op}$ with its multiplication, by reversal of composition. Yoneda gives its stated Morita presentation. Compose the opposite of the proved geometric reconstruction with Proposition 2.4; this realizes precisely the supported category independently specified there by exceptional restrictions. It sends the generator to $D_{S_J} G_{J,S}$ and identifies module mapping spectra by

$$\text{Map}(D_{S_J} M, D_{S_J} N) \simeq \text{Map}(N, M).$$

These maps are induced by the actual duality functor, so their composition is the opposite composition and applying them twice is the evaluation biduality. At the coarse model the generator is explicitly

$$G_S^D = Rj_* \mathbf{Q}_{\ell, U}(1)[2] \oplus \bigoplus_{q \in Z_S} i_{q,*} D\mathbf{Q}_{\ell} A_q.$$

Its closed restrictions need not be those of the star generator. The global pairing uses $R\Gamma_c$ on the primal side; no replacement by $R\Gamma$ on the nonproper open stratum is made. $\square$

At the coefficient level the cellular duality in Morita coordinates is $X \mapsto R \operatorname{Hom}_{\mathbf{B}_S}(X, \mathbf{B}_S)$. Indeed the geometric mapping object $R \operatorname{Hom}(DG_S, DM)$ is $R \operatorname{Hom}(M, G_S)$, and Yoneda identifies the latter with the displayed relative Hom. This establishes the coordinate formula without identifying geometric Verdier duality with a scalar dual of the Morita module.

Definition 5.7 (The cellular Morita target). Its objects are the independently constructed $\mathcal{E}_B(S)$ and their dual types. Its Hom category from an algebra B to an algebra C is the full stable category of (B, C) -bimodules whose underlying right C -module is perfect. Composition is relative tensor in the source-to-target order, and the unit is the regular bimodule. Use the actual module mapping spectra, not their groupoids.

This composition stays in the stated Hom category: for M right C -perfect and L right D -perfect, the exact functor $- \otimes_C L$ sends the thick closure of C into the thick closure of L , hence into $\operatorname{Perf}(D)$. Units and tensor associativity give the enhanced composition cells. No bimodule-smoothness assumption has entered. This also defines the analogous Hom categories on the opposite-algebra coefficient types.

The duality of exact functors between perfect-module categories induces on these Hom categories the relative dual

$$M \mapsto M^D = R \operatorname{Hom}_C(M, C), \quad (B, C) \mapsto (B^{op}, C^{op}).$$

It is right C^{op} -perfect by right C -perfectness of M . To see its composition law without an unjustified scalar-dual formula, use the elementary Yoneda argument for exact functors: a $\mathbf{Q}_\ell$ -linear exact functor $\operatorname{Perf}(B) \rightarrow \operatorname{Perf}(C)$ is tensor with its value on B , carrying the induced left action. Evaluation gives this comparison on the generator, and finite cones and retracts give it everywhere, also on natural transformations. Conjugating such a functor by $X \mapsto R \operatorname{Hom}_B(X, B)$ and its C analogue gives the bimodule M^D , with the displayed endpoint order. Composition of these conjugations and evaluation biduality give its coherent composition and double-dual cells. This constructs D_{cell} in the target. Its identification with geometric kernel mates is proved in Proposition 7.2.

Lemma 5.8 (Relative dual as an opposite bimodule functor). *For ${}_B M_C$, first form $M^\vee = R \operatorname{Hom}_C(M, C)$ with its left C -action by output multiplication and right B -action by precomposition. Its use as ${}_{B^{op}} M_{C^{op}}^D$ means exchange of these two actions via graded symmetry. This defines a contravariant functor on full bimodule mapping categories. For right-target-perfect M, L there is a canonical equivalence*

$$\nu_{L, M} : M^D \otimes_{C^{op}} L^D \xrightarrow{\sim} (M \otimes_C L)^D. \quad (5)$$

It respects evaluation, composition, the regular units and biduality.

Proof. Before exchanging sides, the map is the tensor–Hom evaluation

$$R \operatorname{Hom}_D(L, D) \otimes_C R \operatorname{Hom}_C(M, C) \longrightarrow R \operatorname{Hom}_D(M \otimes_C L, D).$$

For degree-zero elements it sends $g \otimes f$ to $(m \otimes l \mapsto g(f(m)l))$; the chain map uses the Koszul symmetry of tensor factors. Balancing proves linearity for both outer actions. It is the regular evaluation for $M = C$, and both sides are exact and preserve retracts in perfect right C -modules. It is consequently an equivalence for the stated M ; this argument is natural in the left B -action as well. Exchange the sides to obtain the displayed map. Constructing it by tensor evaluation on relative bars makes it natural on the entire mapping categories. For three factors both maps evaluate $h(g(f(m)l)n)$ with the same graded symmetry; the ambient associativity cells identify these evaluations. Units and double duals use the same evaluation. Thus these are the constraints of the exact-functor conjugation already described, with their specific maps now identified. $\square$

For a generated geometric kernel its image is now unambiguously typed:

$$\mathbb{B}_S M_P \mathbb{B}_T = R \operatorname{Hom}_{H_T}(G_T, \Phi_P^! G_S).$$

Preservation and Proposition 5.4 make it right-target-perfect. The geometric evaluation/Fubini comparison $M_P \otimes_{\mathbb{B}_T} M_L \rightarrow M_{L \star_! P}$ is proved separately in Proposition 7.1; it is not part of the definition of the cellular target.

6 Actual Verdier mates and the supported kernel action

Theorem 4.3 puts the whole generated shriek source in the bounded domain of Definition 2.7. Consequently its geometric dual labels and all their actual maps now form the corresponding supported source: composition is $D(L \star_! P)$, with the cells obtained from the already constructed ambient composition and actual biduality. This defines its labels; the next theorem identifies their action by a geometric formula.

Theorem 6.1 (The geometric Hom–shriek–pullback mate). *For a generated kernel $P : a \rightarrow b$ and $N \in \mathcal{A}_a^D$, set*

$$\Psi_{PD}(N) = R p_{b,*} R \operatorname{Hom}_{p_a^* H_a}(P, p_a^! N).$$

This is a bounded constructible object of $\mathcal{A}_b^D$ with its contragredient target action, and the actual evaluation and trace maps give a natural equivalence

$$D_b \Phi_P^!(M) \xrightarrow{\sim} \Psi_{PD}(D_a M) \quad (M \in \mathcal{A}_a). \quad (6)$$

It is natural in coefficient and kernel mapping spectra. No coefficient-dualizability of an attached module is required.

Proof. Write $X = S_a \times S_b$, $A = p_a^* H_a$, and $K = p_a^* M \otimes_A^L P$. The stronger integrand assertion in Theorem 4.3 says that K is bounded constructible. Geometric support duality and tensor–Hom adjunction therefore give the actual comparisons

$$\begin{aligned} D_b R p_{b,!} K &\simeq R p_{b,*} D_X K \\ &\simeq R p_{b,*} R \operatorname{Hom}_A(P, D_X p_a^* M) \\ &\simeq R p_{b,*} R \operatorname{Hom}_A(P, p_a^! D_a M). \end{aligned}$$

In the middle step the right A -action of $p_a^* M$ becomes the left A -action of its dual; the right target action on P becomes the opposite target action on Hom. Tensor–Hom adjunction here is the relative bar/Hom adjunction in module sheaves, using Lemma 2.1 for the bounded module data and the geometric adjoints. It does not replace P or M by a tensor dual. The last equivalence is the ordinary Verdier exchange $D_X p_a^* = p_a^! D_a$ on constructible coefficients.

More explicitly the Hom expression evaluates against K to ω_X . Push its two arguments by $p_{b,!}$ and $p_{b,*}$, use the support projection pairing, and follow by $R p_{b,!} \omega_X \rightarrow \omega_b$. This pairing is the adjoint of the displayed duality map and specifies which comparison is used. Thus the construction is the geometric trace mate, not an isomorphism chosen between two dual objects. Module linearity follows by moving the target action between the evaluation factors.

By preservation $\Phi_P^!(M)$ lies in $\mathcal{A}_b$, so its Verdier dual lies in $\mathcal{A}_b^D$. Every N in the source supported type is canonically $D_a M$ with $M = D_a N$, by geometric biduality. This proves boundedness and the asserted selected target for the already defined Hom expression. It also shows that the Hom integrand is $D_X K$, hence bounded constructible. All three comparisons are natural adjunction maps in the enhancement, proving naturality in all maps of kernels and coefficients. $\square$

Proposition 6.2 (Units, convolution and localization of the mates). *The comparison (6) identifies the star unit with the identity and gives the composition comparison*

$$\Psi_{D(L\star_!P)} \simeq \Psi_{L^D} \Psi_{P^D}.$$

For a coarse open/closed decomposition it takes the actual localization boundary to the boundary of

$$i_* i^! N \longrightarrow N \longrightarrow Rj_* j^* N \longrightarrow i_* i^! N[1]$$

on the supported types, including its compatibility with inserting another generated kernel before or after the triangle.

Proof. For the unit the shriek action is the identity by its geometric graph calculation. The evaluation pairing in the preceding proof is then the usual Verdier evaluation, giving the identity supported action with its actual unit map. For convolution use the common integrand on the threefold product. Adjunction identifies both proposed mate maps with the pairing that evaluates the two kernel factors and then uses the composite support trace. Tensor associativity, Fubini and transitivity of this trace identify these pairings before taking adjoints. Thus their adjoints agree as natural maps. Equivalently, under the specified geometric equivalence they are the composite

$$D_c \Phi_L^! D_b D_b \Phi_P^! D_a \simeq D_c \Phi_L^! \Phi_P^! D_a \simeq D_c \Phi_{L\star_!P}^! D_a,$$

where the middle cancellation is geometric biduality. The preceding pairing calculation is what identifies this expression with the actual Hom-shriek-pullback operations.

The diagonal localization kernels act by $j_! j^*$, identity and $i_* i^*$. Naturality of the same mate on their actual kernel arrows and geometric Verdier exchange give $Rj_* j^*$, identity and $i_* i^!$ in reversed order. Fix the boundary convention as follows: dualize the original cofiber sequence as a fiber sequence and use its canonical stable rotation. If its boundary is $\delta : C \rightarrow A[1]$, the reversed boundary is obtained by suspending $D\delta : D(A[1]) \simeq DA[-1] \rightarrow DC$ with the canonical dual-shift identification. In cochain models the internal-Hom differential is $d(f) = d_{\text{target}} f - (-1)^{|f|} f d_{\text{source}}$; its induced dual-shift identification is used in this rotation. Thus the differential and shift signs are retained, with no independently chosen boundary sign. It agrees with the pullback description of the boundary in Proposition 5.3. Finally convolution is exact in each variable and the mate is a natural map of enhanced functors. Applying it to the arrow/cofiber category, before or after such an exact convolution, yields the same boundary comparison. This proves the asserted mixed localization compatibility, not just equality of its three object values. $\square$

Example 6.3 (Both rectangular convolutions and their Verdier mates). For a rectangle $P_{S,T} = H_S \boxtimes H_T$, its actual supported action is

$$\Psi_{P_{S,T}^D}(N) \simeq R\Gamma(S, N) \otimes_{\mathbf{Q}_\ell} D_T H_T.$$

Indeed apply Theorem 6.1 to the already computed rectangle action. Its scalar compact-support complex is perfect, and global Verdier duality gives $D_{\mathbf{Q}_\ell} R\Gamma_c(S, D_S N) \simeq R\Gamma(S, N)$. The resulting comparison is induced by the same evaluation, so it retains the actions and trace. For two rectangles the shriek middle factor is $C_T = R\Gamma_c(T, H_T)$; the supported middle factor is

$$R\Gamma(T, D_T H_T) \simeq D_{\mathbf{Q}_\ell} C_T.$$

For $U \rightarrow q \rightarrow U$ these factors are A_q and $D_{\mathbf{Q}_\ell} A_q$. For $q \rightarrow U \rightarrow q$ they are $R\Gamma_c(U, \mathbf{Q}_\ell)$ and $R\Gamma(U, \mathbf{Q}_\ell(1)[2])$; the latter has cohomology $\mathbf{Q}_\ell(1)$ in degree -2 and $\mathbf{Q}_\ell$ in degree -1 .

These are actual compact-support and ordinary complexes, respectively. The scalar dual in this calculation must not be confused with the relative algebra dual of a Morita bimodule. Proposition 6.2 applies to the localization triangle convolved with either of these rectangles, thus connects these non-diagonal calculations to the actual boundary maps.

7 Kernel realization in Morita coordinates

For $a = (J, S)$ use $G_a = G_{J,S}$ from Proposition 5.5 and the specified equivalence $B_S \simeq R\mathrm{Hom}(G_a, G_a)$. This uses the independently constructed cellular algebra as a coordinate algebra. It does not redefine the cellular category as the image of a kernel functor. The passage from exact functors on these generator categories to bimodule coordinates is the derived Morita mechanism of [5]; below we specify its evaluation maps and verify their compatibility with the actual geometric convolution.

Proposition 7.1 (Unlocalized enhanced kernel realization). *The actual generated shriek source admits a $\mathbf{Q}_\ell$ -linear enhanced realization in the right-target-perfect cellular Morita target, with kernel image*

$$B_S M_P B_T = R\mathrm{Hom}_{H_b}(G_b, \Phi_P^! G_a).$$

It preserves actual kernel maps and has canonical equivalences

$$M_P \otimes_{B_T}^L M_L \xrightarrow{\sim} M_{L \star_! P}, \quad M_{\Delta_* H_a} \simeq B_S. \quad (7)$$

These are its enhanced unit and composition constraints; they satisfy the associativity and unit cells and retain localization boundaries. No full faithfulness of the kernel-to-action functor is asserted.

Proof. Theorem 4.3 makes each $\Phi_P^!$ an exact functor on the selected categories. Its value on G_a belongs to $\mathcal{A}_b$, so its Morita coordinate is right B_T -perfect. The source generator endomorphisms act through $\Phi_P^!$ and the target ones by precomposition, giving the two commuting actions. Applying the actual integral-transform functor to a kernel map and then applying this mapping functor defines the image on the entire kernel mapping category, including its higher simplices.

Here is the composition map. For $X \in \mathcal{A}_b$, geometric Morita evaluation gives

$$R\mathrm{Hom}(G_b, X) \otimes_{B_T} G_b \longrightarrow X.$$

Apply $\Phi_L^!$ and map from G_c . This constructs a natural map

$$R\mathrm{Hom}(G_b, X) \otimes_{B_T} R\mathrm{Hom}(G_c, \Phi_L^! G_b) \longrightarrow R\mathrm{Hom}(G_c, \Phi_L^! X).$$

It is an equivalence for $X = G_b$, with the regular-module unit map. Both sides are exact in X and preserve retracts, so geometric generation proves it for all X in the selected category. With $X = \Phi_P^! G_a$, compose it with the actual Fubini equivalence $\Phi_L^! \Phi_P^! \simeq \Phi_{L \star_! P}^!$ to obtain (7). Naturality makes it source-linear and natural in both kernel variables. The unit calculation is the actual diagonal action on G_a , hence the regular bimodule, not a supported dualizing module.

To specify coherence rather than choose objectwise associators, first take the enhanced action of ambient kernels on their module categories from Proposition 2.6; the preservation theorem restricts it to the selected full categories. The Morita coordinate equivalences identify their exact functor categories with the right-target-perfect bimodule categories: the comparison is evaluation on the generator, and the same finite-cell Yoneda argument extends

it on natural transformations and their mapping spectra. Transport the actual composition diagram through these equivalences. The evaluation construction just given is precisely its composition map, so the resulting associativity and unit cells belong to the displayed comparison. They are not chosen after checking objects. Exactness sends the specified kernel localization cofiber sequence, with its connecting arrow, to the corresponding bimodule sequence. This establishes the claimed boundary compatibility too. $\square$

Proposition 7.2 (The Morita coordinate of the geometric star mate). *On the supported source the geometric Hom-shriek-pullback action has bimodule coordinate*

$$M_{PD}^* \simeq R \operatorname{Hom}_{\mathbf{B}_T}(M_P, \mathbf{B}_T)$$

with left $\mathbf{B}_S^{op}$ and right $\mathbf{B}_T^{op}$ actions. This comparison intertwines units, convolution, localization and biduality with the cellular duality, before refinement localization.

Proof. Use the actual supported generators $D_a G_a, D_b G_b$. Theorem 6.1 and geometric biduality give

$$\begin{aligned} M_{PD}^* &= R \operatorname{Hom}(D_b G_b, \Psi_{PD}(D_a G_a)) \\ &\simeq R \operatorname{Hom}(D_b G_b, D_b \Phi_P^! G_a) \\ &\simeq R \operatorname{Hom}(\Phi_P^! G_a, G_b) \\ &\simeq R \operatorname{Hom}_{\mathbf{B}_T}(M_P, \mathbf{B}_T). \end{aligned}$$

The third line is induced by the geometric duality functor on mapping spectra; the last is the proved Morita equivalence on its two arguments. Thus the comparison preserves both opposite actions and all kernel maps. Right-target-perfectness gives the relative algebra biduality, and these same maps identify it with geometric evaluation biduality. It is not a scalar-dual calculation.

The geometric unit and convolution comparisons are those of Proposition 6.2; their Morita coordinates are the evaluation comparisons of Proposition 7.1. Both are obtained by applying the same enhanced mapping functors to the specified evaluations, so their unit and composition diagrams commute. The cellular duality on bimodules was constructed from exact-functor conjugation; the displayed calculation identifies that conjugation with the already established geometric mate. Applying this identification to the functor categories of arrows and cofiber sequences gives the localization comparison with its rotated boundary. Applying it twice gives the biduality diagram. This constructs the unlocalized shriek/star Morita compatibility on the whole generated source. Actual product descent and its mixed cells are proved in Sections 8–9.

Writing $\chi_P : M_{PD}^* \rightarrow M_P^D$ for this comparison, its composition square is

$$\begin{array}{ccc} M_{PD}^* \otimes_{\mathbf{B}_T^{op}} M_{LD}^* & \xrightarrow{\mu_*} & M_{D(L \star P)}^* \\ \chi_P \otimes \chi_L \downarrow & & \downarrow \chi_{L \star P} \\ M_P^D \otimes_{\mathbf{B}_T^{op}} M_L^D & \xrightarrow{(\mu_!^D)^{-1} \nu_{L,P}} & M_{L \star P}^D \end{array} \quad (8)$$

To verify the square, evaluate its two adjoints on $G_c, \Phi_L \Phi_P G_a$ and their Verdier partners. The upper route is geometric evaluation on the threefold product followed by the support trace. By the definition of $\mu_!$ as Morita evaluation followed by geometric Fubini, the lower route is the same pairing, with the two regular intermediate factors contracted in the order of Lemma 5.8. Associativity of evaluation and the support projection formula identify the maps before adjunction. The calculation takes place on the multibar diagrams and is natural in their maps, hence gives the specified square on mapping categories, not only on objects. $\square$

8 Proper product descent of the generated kernels

We use the actual kernel categories, not a comparison reconstructed from equality of their actions. In this section I is any cut model below both endpoints under consideration. All supports remain full inverse images of strata unions at the original anchor B ; using I as an intermediate model does not enlarge the support set. For $a = (J, S)$, $I \subset J$, write $a_I = (I, S)$, $u_a : S_J \rightarrow S_I$, and use the following two ambient kernels:

$$U_a = \Gamma_{u_a}^t H_{S_J} : a_I \rightarrow a, \quad V_a = \Gamma_{u_a, *} H_{S_J} : a \rightarrow a_I.$$

Only U_a belongs to the specified refinement generating list. The second kernel is used in the ambient category to construct maps; it is not declared to belong to the generated bounded convolution source or to be an inverse there.

Lemma 8.1 (Ambient graph adjunction). *These kernels have actual unit and counit maps*

$$\eta_a : \Delta_* H_{S_I} \longrightarrow V_a \star_! U_a, \quad \epsilon_a : U_a \star_! V_a \longrightarrow \Delta_* H_{S_J}$$

with the enhanced adjunction triangle cells. The unit is an equivalence on the full inverse-image supports in this paper. No assertion that the counit is an equivalence of ambient kernels is made.

Proof. Graph base change and the regular-module tensor identify $V_a \star_! U_a$ with $\Delta_* Ru_{a, *} H_{S_J}$; properness makes its shriek and ordinary images equal. The unit is the actual nearby algebra unit $H_{S_I} \rightarrow Ru_{a, *} H_{S_J}$, already an equivalence on these complete supports. The opposite graph composite is supported on $S_J \times_{S_I} S_J$. Its coefficient is the relative tensor of the two pulled-back H_{S_J} factors over H_{S_I} , formed in the ambient category. Restricting to its diagonal and multiplying gives its map to $\Delta_* H_{S_J}$ by the diagonal adjunction; this defines ϵ_a . It need not have a bounded source.

To check the triangles at the kernel level, whisker these maps with the graph kernels and use graph base change. The two composites are the unit/counit composites for $u_a^* \dashv Ru_{a, *}$ followed by the regular-module multiplication and its unit. Their triangle and unit identities identify the composites with the identity maps on the graph coefficients; pushing along the graph gives the kernel identities. The same constructions on the relative bar diagrams give their higher cells. This verification does not infer a kernel identity from a possibly nonfaithful action on selected coefficients. Their actions are the actual adjunction $u_{a, H}^* \dashv Ru_{a, *}$, with the specified algebra restriction. $\square$

For $P : a \rightarrow b$ define actual product direct image, restricting the outer actions along the nearby algebra maps, by

$$T_I(P) = R(u_a \times u_b)_* P.$$

Graph base change identifies it, including maps, with

$$T_I(P) \simeq V_b \star_! P \star_! U_a. \tag{9}$$

Indeed the first graph inserts the source proper pushforward variable, and the last one the target proper pushforward variable; their regular coefficient factors cancel. Composition of proper direct images is precisely product direct image. This formula is valid in the ambient categories; it does not require that V_a itself be a selected kernel.

Lemma 8.2 (Direct image of the controlling kernel families). *For $I \subset J \subset K$ the actual comparison maps give*

$$T_I G_{J, K}^{S, T}(W) \simeq G_{I, I}^{S, T}(W), \quad T_I R_{J, K}^{S, T}(W, Z) \simeq R_{I, I}^{S, T}(W, Z).$$

They retain the outer actions and, for localization kernels, their actual arrows and boundary. Product direct image therefore sends the controlling thick family $\mathcal{N}$ into its version at I .

Proof. On a graph, composition of its immersion with $u_a \times u_b$ is the diagonal immersion of W_I followed by $W_K \rightarrow W_I$. Composition of shriek images and properness identify its image with $\Delta_{W_I,!} Ru_* H_{W_K}$. The inverse of the actual algebra unit identifies this with $G_{I,I}(W)$. On a rectangle, proper product base change and projection give $Ru_{a,*} E_S^J(W) \boxtimes Ru_{b,*} E_T^K(Z)$. Proper base change for each complete support square and the algebra unit identify these with $E_S^I(W)$ and $E_T^I(Z)$. These are module comparisons, not just comparisons of scalar cohomology. For a localization triangle the three maps use the same support squares and their units, so exact direct image compares the entire cofiber sequence, including its boundary. Exactness, perfect scalar linearity and preservation of retracts then give the thick-family assertion on objects and all its actual maps. $\square$

Theorem 8.3 (One-sided kernel descent and strong composition). *For every generated kernel $P : a \rightarrow b$, the actual counit gives an equivalence of kernels from a_I to b ,*

$$\alpha_P : U_b \star! T_I(P) = U_b \star! V_b \star! P \star! U_a \xrightarrow{\epsilon_b \star! 1} P \star! U_a. \quad (10)$$

Product direct image restricts to the generated kernel sources and has invertible unit and composition maps

$$\lambda_{L,P} : T_I(L) \star! T_I(P) \longrightarrow T_I(L \star! P). \quad (11)$$

These maps are natural on full kernel mapping spectra and give an enhanced unital composition-preserving functor to the source at I .

Proof. We first prove (10) on the controlling family, not by assuming descent of arbitrary kernels. If $P = G_{J,K}(W)$, its right side is $G_{I,K}(W)$ by the graph convolution formula; its left side is the same graph by Lemma 8.2. Under these specified identifications the map on the graph coefficient is the counit on $H_{W_K} = u_{W,H}^* H_{W_I}$. It is the inverse of the pulled-back algebra unit by the adjunction triangle. Thus it is the claimed equivalence, not an undetermined endomorphism of that graph. For $P = R_{J,K}(W, Z)$, the rectangle-graph calculation on the source side uses the proper algebra unit and gives $R_{I,K}(W, Z)$. On the target side the counit is the one on $E_T^K(Z) = u_{b,H}^* E_T^I(Z)$, with its actual support base change. The unit on $E_T^I(Z)$ is, by proper support base change, the extension by zero of $H_{Z_I} \rightarrow Ru_{Z,*} H_{Z_K}$, hence is invertible. To identify the map itself, write $F_b = u_{b,H}^*$, $R_b = Ru_{b,*}$, $E = E_T^I(Z)$ and $C = E_S^I(W)$. Proper support base change identifies $F_b E = E_T^K(Z)$ and identifies the adjunction unit $\theta_E : E \rightarrow R_b F_b E$ with the extension of the algebra unit on Z_I . Graph base change and regular-module cancellation give the following square of kernels on $S_I \times T_K$:

$$\begin{array}{ccc} U_b \star T_I(P) & \xrightarrow{\alpha_P} & P \star U_a \\ \beta_{\text{left}} \downarrow \simeq & & \downarrow \beta_{\text{right}} \\ C \boxtimes F_b R_b F_b E & \xrightarrow{1 \boxtimes \epsilon_{F_b E}} & C \boxtimes F_b E. \end{array} \quad (12)$$

The right vertical map is also an equivalence. The vertical maps use the same source algebra-unit inverse as (3); on the target they use only proper support base change. Under these maps the top arrow, defined by the graph counit, is the direct-image evaluation followed by the H_b -action, exactly the definition of $\epsilon_{F_b E}$. Naturality of base change with evaluation proves the square before any counit is inverted. Moreover the adjunction triangle is

$$\epsilon_{F_b E} \circ F_b(\theta_E) = 1_{F_b E}.$$

Since θ_E is the specified equivalence, the bottom arrow has inverse $1 \boxtimes F_b(\theta_E)$. This proves invertibility of this particular α_P , not just isomorphism of its endpoints. All maps are linear for the two outer actions. This proves the assertion on both families as an actual kernel map.

Both sides of (10) are exact in P and preserve retracts, even though the ambient counit need not itself be invertible. Hence the assertion extends to $\mathcal{N}$ and in particular to the whole generated source.

Using (9), define (11) by inserting the middle counit:

$$V_c \star L \star U_b \star V_b \star P \star U_a \longrightarrow V_c \star L \star P \star U_a.$$

It is $V_c \star L$ applied to the just proved α_P , and is therefore an equivalence. The unit comparison is η_a of Lemma 8.1. For three kernels the two composition maps insert counits at the same two middle places. The interchange law identifies their order of insertion, with its cell in the ambient enhancement. The adjunction triangles give the unit cells. Iterated insertion on relative bar diagrams supplies all higher substitution cells. This proves coherence for these actual maps, rather than selecting abstract associators afterwards.

In particular the compatibility needed for localization is the following specified homotopy, with ambient associators suppressed:

$$\alpha_{L \star P} \circ (U_c \star \lambda_{L,P}) = (L \star \alpha_P) \circ (\alpha_L \star T_I(P)). \quad (13)$$

Both sides start at $U_c \star T_I(L) \star T_I(P)$ and end at $L \star P \star U_a$. In the sandwich expressions they remove the same $U_c V_c$ and $U_b V_b$ factors; the ambient interchange homotopy identifies the two orders. The unit identity is

$$\alpha_{\text{id}_a} \circ (U_a \star \eta_a) = \text{id}_{U_a},$$

which is the graph adjunction triangle. These are homotopies between the specified kernel maps, natural on their entire mapping spectra, not merely equalities of their induced coefficient actions.

It remains to check that the codomain is the specified generated source, not merely $\mathcal{N}$. On every original generator the preceding direct-image calculation gives its corresponding generator at I : inclusion graphs, units, localization terms and the two rectangles; a refinement graph gives the identity kernel on its support at I . Exactness preserves finite cones, twists and retracts, and the now proved invertible λ preserves every finite convolution word. Since the generated subcategories are full, images of their arbitrary kernel maps also belong to them. This proves the claimed restriction without changing the generating list or imposing a preservation condition in its definition. $\square$

Remark 8.4 (What has and has not descended). The key equivalence is after source whiskering by U_a . It does not identify a fine kernel with ordinary two-sided pullback of its coarse image. In particular $(u \times u)^{-1} \Delta_S = S' \times_S S'$ still need not equal $\Delta_{S'}$. For the diagonal we used proper pushforward and then a one-sided graph pullback. No equivalence of unrestricted product kernel categories has been proved or used.

Proposition 8.5 (The actual four-model diamond). *The two product-direct-image paths from $\{1, 2, 3\}$ through $\{1, 2\}$ and through $\{2, 3\}$ to $B = \{2\}$ have the same coherent comparison with direct product pushforward to B . This holds on kernels and their maps, and respects α , λ and all coarse support localization diagrams.*

Proof. For a chain $I \subset J \subset K$, proper composition gives

$$R(u_{JI,S} \times u_{JI,T})_* R(u_{KJ,S} \times u_{KJ,T})_* \simeq R(u_{KI,S} \times u_{KI,T})_*.$$

The restricted outer algebra actions agree because the actual nearby pullback algebra maps compose, not merely because their cohomology algebras are isomorphic. On graph coefficients the comparison is $H_{W_I} \rightarrow Ru_{JI,*} H_{W_J} \rightarrow Ru_{KI,*} H_{W_K}$, which is the direct algebra

unit. On rectangles it is the external product of the two such comparisons; proper product base change and its naturality identify this with the displayed composition map. On all kernels we use this actual natural transformation of proper functors, so it is already defined on arbitrary maps and higher cells.

The graph units and counits for a composite are the composite adjunction units and counits, by their explicit construction. Whiskering these identities in (10) and (11) proves compatibility with α and λ , including when a rectangle is convolved with a graph or a localization kernel. Applying the exact comparison to the arrow and cofiber-sequence categories gives the boundary comparison. The integer-cut chart maps in [2] identify the two composites in the stated diamond as the same actual scheme map. Composition in the fixed enhancement therefore gives both paths one comparison with its direct image, with the associativity cells for longer strings. This is a comparison on actual products; the single-support module equivalence alone was not used as its proof. $\square$

9 Verdier, refinement and localization in one diagram

Proposition 9.1 (Actual actions and proper Verdier product comparison). *Write $F_a = u_{a,H}^*$, $R_a = Ru_{a,*}$ on the selected coefficient types. The product comparison has actual action*

$$\Phi_{T_I(P)}^! \simeq R_b \Phi_P^! F_a, \quad F_b \Phi_{T_I(P)}^! \xrightarrow{\sim} \Phi_P^! F_a.$$

The second map is the action of α_P . Define $T_I^ : \mathbb{K}_*(a^D, b^D) \rightarrow \mathbb{K}_*((a_I)^D, (b_I)^D)$ on labels, module maps and higher simplices by actual $R(u_a \times u_b)_*$, restricting both opposite algebra actions along the nearby algebra units. It has the proper-duality equivalence*

$$T_I^*(P^D) = R(u_a \times u_b)_* P^D \simeq D_{a_I \times b_I} T_I(P). \quad (14)$$

These identifications agree with the actual Hom-shriek-pullback star actions, including their evaluations and biduality.

Proof. The first action formula follows either by applying the actual graph actions to (9), or directly by proper product base change and projection. The former calculation identifies the map as the very same composite of the units and counits, so the second map is precisely the action of (10). This also gives, using the selected counit $F_a R_a \simeq \text{id}$, $\Phi_{T_I(P)}^! R_a \simeq R_b \Phi_P^!$.

Put $\pi = u_a \times u_b$. This product map is proper, and P is bounded constructible by the kernel-preservation theorem. Proper Verdier duality applies to it. Its comparison is adjoint to the pairing

$$R\pi_* P \otimes R\pi_* DP \longrightarrow R\pi_*(P \otimes DP) \longrightarrow R\pi_* \omega_{a \times b} \xrightarrow{\text{Tr}_\pi} \omega_{a_I \times b_I}.$$

Moving either outer algebra action between the two evaluation arguments proves its opposite-action linearity, so this is an identification of kernel modules with their actual maps. It defines T_I^* by actual product direct image, independently of using the displayed equivalence as a description of its value.

On coefficient types the supported transition is $F_a^D = D_a F_a D_{a_I}$, with inverse $\bar{R}_a = D_{a_I} R_a D_a : \mathcal{A}_a^D \rightarrow \mathcal{A}_{a_I}^D$. Biduality and the selected identities $F_a R_a \simeq 1$, $R_a F_a \simeq 1$ give $F_a^D \bar{R}_a \simeq 1$ and $\bar{R}_a F_a^D \simeq 1$. Proper duality identifies $\bar{R}_a$ with actual $Ru_{a,*}$ on the opposite-action supported modules; this is a separate restriction of the geometric functor, not the primal functor with an unchanged domain. The actual star formula was identified with this conjugation in Theorem 6.1. Its evaluation pairing, after product pushforward, is the pairing above followed by the target projection trace. Performing instead the projection first and then u_b gives the same pairing: both are evaluation followed by the trace of the

same composite proper-support map. Projection base change and trace transitivity identify these maps before taking adjoints. Thus (14) and the supported action comparison are the mates of the shriek product comparison, not arbitrary equivalences with its abstract dual. Applying this argument twice and using evaluation gives the biduality square.

For clarity, write $\chi_{\pi,P} : R\pi_* DP \rightarrow DR\pi_* P$ for this comparison. If $\rho : X' \rightarrow X$ and $\pi : X \rightarrow Y$ are proper, its refinement transitivity is the specified homotopy

$$\chi_{\pi\rho,P} = \chi_{\pi,R\rho_*P} \circ R\pi_*\chi_{\rho,P}, \quad (15)$$

under the proper-composition identifications. Both adjoint pairings are evaluation followed by $\mathrm{Tr}_\pi \circ R\pi_* \mathrm{Tr}_\rho = \mathrm{Tr}_{\pi\rho}$. This explicitly identifies the Verdier comparison along the two refinement paths. Inserting a convolution uses the same identity for support traces, with $!$ on nonproper projections, and the projection pairing; it does not replace those projections by proper maps. $\square$

Proposition 9.2 (The product, convolution and boundary comparison diagrams). *The proper product comparison, the convolution constraint λ and their Verdier mates are comparisons of enhanced functors. They commute with the specified localization arrows and their connecting maps, including along the four-model diamond.*

Proof. Here are the geometric squares used in the comparison; their Cartesian status matters. For contractions u_a, u_b, u_c , put $X = a \times b \times c$, $X_I = a_I \times b_I \times c_I$ and $Y = a_I \times b \times c_I$. With the middle projections denoted p, p_Y, p_I , the square

$$\begin{array}{ccc} X & \xrightarrow{s=u_a \times 1 \times u_c} & Y \\ p \downarrow & & \downarrow p_Y \\ a \times c & \xrightarrow{v=u_a \times u_c} & a_I \times c_I \end{array}$$

is Cartesian, and s, v are proper. The actual shriek Beck–Chevalley map is $v^* R p_{Y,!} \rightarrow R p_{I,!} s^*$. Its projection-formula mates and the composite identity $p_Y s = v p$ give the comparison $R p_{Y,!} R s_* \simeq R v_* R p_{I,!}$. The additional map $\rho = 1 \times u_b \times 1 : Y \rightarrow X_I$ is proper; its square with p_Y, p_I and the identity of $a_I \times c_I$ is *not* asserted Cartesian. Instead its contribution to composition is the middle graph adjunction counit.

More explicitly its support before that counit is $b \times_{b_I} b$, with diagonal $\delta_b : b \rightarrow b \times_{b_I} b$. Its two coefficient factors tensor over the pulled-back coarse algebra. Restriction along δ_b and coefficient multiplication are exactly $\epsilon_b : U_b V_b \rightarrow 1$. This supplies the non-Cartesian middle face. Adjoin to the displayed square the support square for δ_b and the projection squares for the two kernel factors. This describes the cube as a pasting of Cartesian Beck–Chevalley faces and the one specified counit face, rather than treating the whole middle face as base change. After relative bar realization, its two shriek routes are

$$V_c L U_b V_b P U_a \xrightarrow{V_c L \epsilon_b P U_a} V_c L P U_a,$$

with the exterior faces given by the displayed proper-product comparison. This is precisely λ , so this diagram identifies the actual map on all kernel morphisms.

For its Verdier mate apply evaluation on the same support product. On each Cartesian face use the mate of the specified Beck–Chevalley map, not an independently chosen isomorphism. On the middle face evaluation is balanced over the coarse algebra; it therefore identifies the counit multiplication with its Hom mate. Finally trace on the two outer routes is identified by

$$\mathrm{Tr}_{p_Y} R p_{Y,!} (\mathrm{Tr}_s) = \mathrm{Tr}_{p_Y s} = \mathrm{Tr}_{vp} = \mathrm{Tr}_v R v_* (\mathrm{Tr}_p).$$

These are maps from the appropriate pushforward of ω_X ; $!$ is retained on p, p_Y , which need not be proper. Projection pairing and trace transitivity in the enhanced six-operation input make this an identity of the specified natural transformations. It is the product/convolution/Verdier mixed square.

For precision about the diamond, write f_{ij} for its actual product contractions and $c_{h,g} : Rh_*Rg_* \rightarrow R(hg)_*$ for the composition cell of the fixed enhancement. Both paths to the anchor are compared to the direct functor by these c 's; the comparison of paths is $c_{\text{right}}^{-1}c_{\text{left}}$, not a claim that chosen cells are literally equal. For three steps the specified simplex is

$$c_{h,gf} \circ (Rh_*c_{g,f}) = c_{hg,f} \circ (c_{h,g}Rf_*).$$

The algebra units are images of the same composition simplex under the nearby functor. On graphs it acts on their actual algebra unit, on rectangles on its external product, and on the middle counit face on the composite adjunction. The trace calculation above identifies its dual simplex. These are natural transformations of the full ambient functors, so restriction to the generated thick subcategories retains the simplices on all their maps.

Finally let $Q_0 \rightarrow Q_1 \rightarrow Q_2 \xrightarrow{d} Q_0[1]$ be a specified localization triangle. Each natural comparison τ above between exact functors F, G induces the square

$$\begin{array}{ccc} FQ_2 & \xrightarrow{F(d)} & FQ_0[1] \\ \tau_{Q_2} \downarrow & & \downarrow \tau_{Q_0[1]} \\ GQ_2 & \xrightarrow{G(d)} & GQ_0[1]. \end{array}$$

It is obtained by applying the same natural transformation to the cofiber diagram of $Q_0 \rightarrow Q_1$. Whiskering before or after that arrow is exact; taking its Verdier mate reverses the triangle with the fixed dual-shift convention. This proves the boundary and diamond compatibility for the actual connecting maps. $\square$

Lemma 9.3 (Assembly from specified enhanced adjunctions and mates). *Work in a fixed enhanced kernel bicategory with exact composition constructed on relative bar diagrams. Suppose given graph adjunctions $U_a \dashv V_a$ and their enhanced unit, counit and triangle cells. On full stable subcategories closed under the specified operations suppose that:*

1. *the graph units and the whiskered counits α_P are invertible, and the proposed T , λ and α are the sandwich and counit-insertion maps, with (13) and the graph unit identity;*
2. *the base-change and refinement comparisons are the actual pasting comparisons, identifying the adjunctions for composite maps with the composite adjunctions;*
3. *the bounded duality has evaluation biduality, and the proposed star and proper-duality comparisons are the mates of the same evaluations and support traces, with projection pairing and transitive trace as in (15).*

Then these specified comparisons assemble into a coherent diagram of composition, units, base change, refinement and duality on those subcategories. Exactness adds their localization cofiber diagrams. Transport by enhanced coefficient Morita equivalences preserves this diagram and its mixed cells.

Proof. Construct the diagram first in the ambient enhanced categories. A string of composable kernels is evaluated by its relative bar multisimplicial diagram. Removing a middle UV uses the given counit; independent removals commute by enhanced interchange, and inserting then removing a unit is its given triangle cell. Thus counit insertion defines the

composition constraints and all their substitutions. The first hypothesis identifies the named maps with these constraints and makes them invertible on the selected subcategory. Pasting and composite adjunctions identify their base-change and refinement faces by the second hypothesis.

Taking mates is the operation of composing with the given units, counits and evaluation on mapping categories. It therefore carries the already constructed simplices and their substitutions to mate simplices. The third hypothesis identifies the named geometric star maps with those mates: on each mixed face both adjoint maps are the same evaluation followed by the same composite trace. Evaluation biduality supplies the double-dual cells. Restriction to full stable subcategories retains these diagrams, since all named operations and values belong to them. Passing to arrow and cofiber categories, and then through enhanced Morita equivalences, gives the last assertions.

This is a restriction and identification lemma, not a claim that finitely many equalities in a homotopy category determine every higher homotopy. The enhanced adjunctions, pasting operations and trace diagrams are part of its input. The mathematical checks in an application are invertibility and identification of the *specified geometric maps* with these enhanced constructions. $\square$

Proposition 9.4 (Joint compatibility on the generated kernel diagram). *Product descent, kernel convolution, their units, coarse localization boundaries, geometric Verdier duality and biduality form a coherent diagram on the generated sources. The Morita realizations carry this diagram to the corresponding cellular one. In particular the new kernel operations and the actual product diamond, rather than only the earlier coefficient operation diagram, are included.*

Proof. We apply Lemma 9.3 to the actual constructions, not to abstract objectwise equivalences. Its first hypothesis is Lemma 8.1 and Theorem 8.3, with the rectangle identification (12). Its second is the specified composition simplex and proper support base-change maps in Proposition 9.2, together with Proposition 8.5. Its third is Theorem 6.1, Proposition 6.2 and Proposition 9.1, with the mixed cube and trace identity of Proposition 9.2. Boundedness and closure are Theorem 4.3. We spell out the mixed geometric identification and its Morita image below.

First work in the fixed ambient enhanced kernel and module categories. All kernel arrows are retained. The shriek part of the diagram is built from the specified convolution functors, proper product direct images, η , ϵ , and their whiskerings. The maps α and λ are those whiskerings, not independent choices. Their substitution identities are the interchange and adjunction identities verified in Theorem 8.3. Composition of proper images with these maps is the composite adjunction identity of Proposition 8.5. Those arguments also identify the elementary geometric maps with these enhanced cells.

Include the dual types and evaluations. Verdier duality here is applied to the bounded kernels and whiskered comparison maps proved above, not to an asserted reflexive structure on the possibly unbounded ambient $U_a \star V_a$. The mate theorem identifies each star action with its actual Hom-shriek-pullback expression. Proposition 9.1 identifies its new proper comparison with the trace mate. For a composite of a convolution square and a product square, both proposed Verdier comparisons are adjoint to evaluation on the same multiple product followed by the same composite trace. Interchange, the projection map and trace transitivity identify these evaluations. This proves the mixed Verdier/convolution/refinement square, not just its separate edges. The same evaluation supplies the unit and biduality squares. The functors in question are exact, with duality contravariantly exact. Apply these specified maps to their full arrow and cofiber categories; their naturality identifies the mixed localization boundary with the dual-shift convention fixed above.

These constructions are made on the relative bar diagrams and the functor categories of strings of actual maps. Whiskering, taking mates and taking cofibers are operations on these enhanced categories, so they send the entire simplices and their substitutions to the corresponding simplices. The unit, interchange and trace identities just checked identify the required geometric generators of this operation diagram. Restricting to the proved bounded generated subcategories therefore retains all its higher mixed cells. We do not claim uniqueness of an arbitrary enhancement of every six operation, nor invoke the old coefficient transport theorem for unverified new operations.

Finally, in generator coordinates the action formula and the actual adjunction $F_b \dashv R_b$ give

$$M_{T_I(P)} = R\mathrm{Hom}(G_{b_I}, R_b \Phi_P^! F_a G_{a_I}) \simeq R\mathrm{Hom}(F_b G_{b_I}, \Phi_P^! G_a) = M_P.$$

Here $F_a G_{a_I} = G_a$ and $F_b G_{b_I} = G_b$ are the already specified geometric generator comparisons. This equivalence is natural on kernel maps and respects both source and target actions. Denote it by σ_P , and denote the evaluation constraint (7) by $\mu_{L,P}$. The compatibility that allows strong convolution to descend is explicitly

$$\sigma_{L \star P} \circ M(\lambda_{L,P}) \circ \mu_{T_I(L), T_I(P)} = \mu_{L,P} \circ (\sigma_P \otimes \sigma_L). \quad (16)$$

Both sides map $M_{T_I(P)} \otimes M_{T_I(L)}$ to $M_{L \star P}$, using the specified identifications of endpoint algebras. Applying the same adjunction to the evaluation map defining Morita composition identifies the two tensor comparisons; its triangle identities give the unit comparison. On the dual generators the proper evaluation just checked gives the relative algebra dual comparison of Proposition 7.2. Hence it also retains biduality and the boundary maps. The full exact-functor/Morita equivalences transport the entire diagram and its higher cells. This last transport is applied only after the new kernel and product maps have been constructed and identified geometrically. $\square$

Theorem 9.5 (Anchored localization). *The two enhanced Morita realizations descend through the specified refinement localizations and satisfy the equivalence in the main theorem, preserving units, convolution, localization, biduality and the actual four-model comparison. More precisely product direct image to B induces an equivalence between the localized generated shriek source and its full subcategory of objects at B , with the generated kernels there. The dual statement holds for the supported source. This is anchored independence for these sources, not for all geometric kernels or for a change of anchor.*

Proof. Let T_B be the enhanced product functor just constructed, and let ι include the kernel source at B . The restricted composite $T_B \iota$ is the identity, with the identity proper maps and their unit cells. Every distinguished refinement graph is sent by T_B to the regular unit kernel on its coarse support. Thus T_B factors through the enhanced localization at $\mathcal{W}_B$.

The components $U_a : \iota T_B(a) \rightarrow a$, together with the invertible kernel maps α_P , give a coherent natural comparison from ιT_B to the identity on the generated source. Its compatibility with composition is precisely (13), and its unit compatibility is the graph triangle identity proved there. These components become equivalences after inverting $\mathcal{W}_B$; for composite refinements the same conclusion follows from their graph composition. Hence ι and the descended T_B are inverse equivalences there. Equivalently these identities verify the localization universal property directly: any enhanced functor inverting the U_a is identified, by its images of the α_P , with its restriction to the anchor composed with T_B . The same comparison on functor mapping categories retains transformations and their higher cells. More explicitly, precomposition with T_B and ι gives functors in the two directions of (1). Their composite on the anchor side is the identity. On a functor F inverting $\mathcal{W}_B$, the other composite has comparison with F given by $F(U_a)$ and $F(\alpha_P)$. For a pseudonatural

transformation $\tau : F \rightarrow F'$ its naturality square at U_a is exactly the modification relating these comparisons. Naturality on modifications gives the same construction on their mapping spaces. Since $F(U_a)$ are equivalences, this is an equivalence of these functor categories, proving the stipulated universal property with the anchor as the explicit localization model. This argument uses the actual maps α_P between generated kernels; it neither quotients kernel maps by equality of action nor inserts a nonselected reverse graph as an a priori inverse.

The shriek Morita functor sends U_a to the bimodule representing F_a , already an equivalence of the selected coefficient categories. It therefore descends by the universal property of localization. The supported functor does the same for the dual graph arrows, which act by the proved geometric F_a^D . The geometric and cellular dualities carry the distinguished arrows to these supported arrows. The enhanced natural equivalence proved before localization, with its entire compatible diagram in Proposition 9.4, therefore descends as well. This proves the main theorem. All product and trace comparisons remain those of the fixed maps; the localization is performed only after their verification. $\square$

Proposition 9.6 (A noninvertible kernel survives localization). *The coarse rectangular kernel $P_{Uq} = H_U \boxtimes A_q$, for either endpoint q , is not invertible in the anchored localized source.*

Proof. Its actual action sends $\mathbf{Q}_{\ell,U}$ to $C \otimes A_q$, where $C = R\Gamma_c(U, \mathbf{Q}_{\ell})$. The source endomorphism complex is $A = R\Gamma(U, \mathbf{Q}_{\ell})$, whose total cohomology dimension is two. Since C is a perfect $\mathbf{Q}_{\ell}$ -complex, the target endomorphism complex is

$$R\mathrm{Hom}_{A_q}(C \otimes A_q, C \otimes A_q) \simeq R\mathrm{Hom}_{\mathbf{Q}_{\ell}}(C, C) \otimes A_q.$$

The cohomology of C is one-dimensional in each of degrees 1, 2, and that of A_q is one-dimensional in degrees 0, 1. Kunneth therefore gives total dimension eight for this endomorphism complex (in degrees $-1, 0, 1, 2$ the dimensions are 1, 3, 3, 1). Thus the action is not fully faithful and cannot be an equivalence. The Morita realization factors through refinement localization by Theorem 9.5. An invertible kernel there would have an invertible Morita image and hence an equivalence as its coefficient action, contradicting this calculation. No faithfulness of the kernel realization is used in this implication. $\square$

The following example uses the standard Hochschild interpretation of identity-functor endomorphisms [5]. Its geometric source consists of constructible ℓ -adic kernels acting on the selected coefficient category. It is not the quasicohherent kernel category in the integral-transform comparisons of [6]. Full bimodule mapping objects in the Morita target can therefore contain maps absent from this geometric source; we compute that difference explicitly rather than importing a quasicohherent kernel–functor equivalence.

Proposition 9.7 (Geometric unit maps and Hochschild mapping-object discrepancy). *On the coarse open stratum $U = \mathbf{G}_m$, put $A = R\Gamma(U, \mathbf{Q}_{\ell})$ and $I_U = \Delta_* \mathbf{Q}_{\ell,U}$. The geometric source and its Morita target have mapping objects*

$$R\mathrm{Hom}_{U \times U}(I_U, I_U) \simeq A, \quad R\mathrm{Hom}_{A^e}(\mathbf{Sk}_!(I_U), \mathbf{Sk}_!(I_U)) \simeq R\mathrm{Hom}_{A^e}(A, A).$$

Their degree-zero cohomology groups are not isomorphic. With the geometric coefficients of this paper, a choice of generator of $H^1(U, \mathbf{Q}_{\ell})$ identifies the second with $\mathbf{Q}_{\ell}[[z]]$, whereas the first is $\mathbf{Q}_{\ell}$. The realization, also after anchored localization, is consequently not fully faithful on kernel mapping objects.

Proof. Here $H_U = \mathbf{Q}_{\ell,U}$, so the geometric module–kernel mapping object is ordinary sheaf Hom. Closed-immersion direct image along Δ is fully faithful, with $\Delta^* \Delta_* \simeq 1$. Its adjunction gives the first equivalence. The unit calculation in Proposition 7.1 gives the regular bimodule

A , so the target, which retains full bimodule mapping spectra, gives the second. Right-target-perfectness does not require A to be perfect over A^e and does not truncate this Hom.

For completeness, compute the latter Hom rather than replacing its totalization by a direct sum. Over the geometric field, choose a basis of the one-dimensional Tate line $H^1(U, \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-1)$. We use characteristic-zero rectification here: commutative dg $\mathbf{Q}_\ell$ -algebras model E_∞ algebras in complexes, since the finite symmetric-group coinvariants are exact over $\mathbf{Q}_\ell$ and the commutative operad is equivalent to its E_∞ resolution. Derived global sections of the constant commutative algebra supply such an E_∞ model. A rectified commutative dg model of A is quasi-isomorphic to $\Lambda(\epsilon)$, $|\epsilon| = 1$: a cocycle representing this basis defines a map from the free graded-commutative algebra on one odd generator, and the known cohomology in degrees 0, 1 makes it a quasi-isomorphism. We use this as a model choice for this calculation only; no Galois-equivariant trivialization of a Tate line is claimed.

Write $A_0 = \Lambda(\epsilon)$ and $A_0^e = \Lambda(\epsilon_l, \epsilon_r)$, with the graded opposite convention. A semifree A_0^e -resolution of its diagonal module is

$$F = A_0^e \otimes \mathbf{Q}_\ell[t], \quad |t| = 0, \quad dt = \epsilon_l - \epsilon_r, \quad F \longrightarrow A_0, \quad t \longmapsto 0.$$

The filtration by polynomial degree is semifree, since the differential lowers that degree. To check the augmentation, use coordinates ϵ_r and $\delta = \epsilon_l - \epsilon_r$. The remaining complex is $\mathbf{Q}_\ell[t] \xrightarrow{\partial_t} \delta \mathbf{Q}_\ell[t]$. In characteristic zero its kernel is the constants and its cokernel is zero, proving the quasi-isomorphism.

Set $t^{[n]} = t^n/n!$. Then $dt^{[n]} = \delta t^{[n-1]}$. On the diagonal module δ acts by zero, whence

$$R\mathrm{Hom}_{A_0^e}(A_0, A_0) \simeq \mathrm{Hom}_{A_0^e}(F, A_0) = \prod_{n \geq 0} A_0,$$

with zero differential. The product is essential: F is a direct sum of free generators all in degree zero, and ordinary derived Hom takes that sum to a product. The resolution diagonal $t \mapsto t \otimes 1 + 1 \otimes t$ gives $t^{[n]} \mapsto \sum_{i+j=n} t^{[i]} \otimes t^{[j]}$. Thus the cup product on degree-zero sequences is the power-series product, giving $H^0 = \mathbf{Q}_\ell[[z]]$. This is the usual derived Hom, not a polynomial or a separately imposed continuous variant.

The degree-zero map from the geometric source sends scalars to scalar multiples of the identity, hence to the constant series. It is not surjective. Finally, Theorem 9.5 identifies the localized source with its full generated anchor source, so it does not add these missing endomorphisms to I_U . The discrepancy persists after localization. It describes additional coherent natural transformations in the Morita action category, and is compatible with all the strong composition assertions above. $\square$

10 Scope of the result

The source is generated by the named geometric nearby kernels and all their actual morphisms. Inverting its refinement graphs gives model independence only for $xy = t^4$, $B = \{2\}$ and the four stated models, with complete inverse-image supports. The two rectangular convolutions and their nontrivial cochain factors are retained in this localization; it is not a category consisting only of invertible operations. The cellular target retains arbitrary coarse attachments and right-target-perfect bimodules, without asserting smoothness of the cellular algebras or full faithfulness of kernel realization.

No original log/root kernel realization, integral coefficient extension, external infinite base inertia, arbitrary fine support, change of anchor or unrestricted product-kernel equivalence is included. These would require additional geometric comparison theorems. The present statement supplies a fixed anchored, nearby-enhanced kernel and Verdier interface; it does not by itself construct the full orbital correspondence or model-independent Grand Theorem.

Appendix A proves only the local completed-root realization, the two coarse rectangular tests and the selected carrier comparison. Appendix B excludes ordinary stalk/pullback gluing of the distribution unit. The actual exceptional mate retains the full supported dualizing complex; it does not construct a global ordinary shriek kernel calculus. That construction is a separate research problem, not an unproved part of Theorem 1.1. In particular the nearby result does not depend on solving it. Proposition 9.7 also rules out promoting the stated Morita realization to a full kernel equivalence on the open stratum. Neither adding models nor changing the log notation supplies the missing root interface.

A Local completed-root realization

This appendix records local results independent of the proof of Theorem 1.1: completed residual-gerbe units, normalized rectangles, the two coarse passages and their selected carrier pairing. None asserts a global four-model root kernel functor. Continuous cohomology and actual root shriek integration remain distinct operations until the specified local normalization is applied.

Fix a node, its relative chart and the compatible base-root neutralization of [1]. Write $\Gamma = \mathbf{Z}_\ell(1)$, $G_n = \Gamma/\ell^n\Gamma$ and $\Lambda_r = \mathbf{Z}/\ell^r$. Relative products here use two independent relative inertia factors *after* this neutralization; they do not retain a common absolute base inertia factor. At finite level they are $BG_n \times BG_n = B(G_n \times G_n)$.

For $\Gamma_a = 0$ or Γ , put $A_{a,r} = R\Gamma_{\text{cts}}(\Gamma_a, \Lambda_r)$ and $I_{a,r} = C_{\text{lc}}(\Gamma_a, \Lambda_r)$. The inflation assembly of the finite permutation diagonals is $C_{\text{lc}}(\Gamma_a, \Lambda_r)$; continuous Shapiro identifies its cochains on Γ_a^2 with $A_{a,r}$, with both actions induced by diagonal restriction. Its scalar dual is the completed distribution module $\varprojlim_n \Lambda_r[G_n]$, with quotient-fibre summation. These statements follow by finite coinduction and by dualizing the filtered union of finite function modules. This cohomological comparison is not a shriek comparison. The following calculation specifies the difference before constructing the local completed shriek realization.

Proposition A.1 (A node passage distinguishes star and trace assembly). *At a node, continuous cohomological integration of the constant coefficient gives $A_r = C_{\text{cts}}^*(\Gamma, \Lambda_r)$. In contrast, assembling the actual finite-stack shriek images along their trace transitions gives*

$$J_r := R\varprojlim_n Rg_{n,!}\Lambda_r \simeq R\text{Hom}_{\Lambda_r}(A_r, \Lambda_r), \quad g_n : BG_n \rightarrow \text{Spec } k.$$

After adic recovery and rationalization, A has cohomology in $(0, 1)$ whereas $J = D_{\mathbf{Q}_\ell}A$ has cohomology in $(-1, 0)$. Consequently trace-assembled shriek convolution of the two constant root rectangles through this node does not give the nearby rectangle convolution factor A .

Proof. At finite precision, [1, prop:frag-finite-quotient-enhancement] identifies the actual geometric shriek adjunction with derived coinvariants, including its counit and composition cells. Its prop:frag-point-geometric-bars identifies the full bar and its geometric pairing. Thus the specified comparison is

$$Rg_{n,!}\Lambda_r \simeq \Lambda_r \otimes_{\Lambda_r[G_n]}^L \Lambda_r \simeq D_{\Lambda_r} C^*(G_n, \Lambda_r).$$

Under this comparison the quotient trace is the dual of inflation, with the coefficient arrows specified by [1, lem:frag-point-trace-diagram-descent]. These are unbounded full bars; BG_n has dimension-zero dualizing coefficient Λ_r . Therefore

$$R\varprojlim_n D_{\Lambda_r} C^*(G_n, \Lambda_r) \simeq D_{\Lambda_r} \text{hocolim}_n C^*(G_n, \Lambda_r) = D_{\Lambda_r} A_r.$$

There is no truncation of the unbounded finite bars in this step. The resulting A_r is the perfect two-term continuous complex; its dual is likewise perfect and compatible with derived precision reduction. Recovering r and then rationalizing gives $H^{-1}(J) = \mathbf{Q}_\ell(1)$, $H^0(J) = \mathbf{Q}_\ell$, whereas $H^0(A) = \mathbf{Q}_\ell$, $H^1(A) = \mathbf{Q}_\ell(-1)$.

For the passage $U \rightarrow q \rightarrow U$, the two constant root rectangles have constant middle coefficient. Finite-stack convolution thus integrates that coefficient by $Rg_{n,!}$. Retaining the trace assembly gives the scalar factor J , multiplying the unchanged outer rectangular coefficient. Near the node the convolution in Proposition 3.1 instead has factor A . An exact $\mathbf{Q}_\ell$ -linear realization that restricts to the identity kernel functor on the root-free open stratum (and hence on its scalar-tensored rectangular kernels), and sends these two constant rectangles to the named nearby rectangles cannot identify these composites. This is an obstruction to this *specific* naive choice of shriek assembly and rectangles, not to every possible normalized root source. The finite-carrier supported trace in [1] has, in addition, its carrier shift and twist; it cannot remove this distinction by being renamed ordinary inertia integration. $\square$

A.1 Completed shriek kernels and the source normalization

We now resolve the node-integration discrepancy for the specified rectangular kernels. The endpoints in the point-valued calculation carry inertia 0 or $\Gamma = \mathbf{Z}_\ell(1)$, as occurs on strata of the fixed models. An endpoint with zero inertia is a point; it is not the actual open scheme $U = \mathbf{G}_m$. No statement for general root stacks is needed. The result is local on these strata. A global kernel calculus is not obtained by gluing these modules through ordinary stalk pullback; the obstruction is proved in Appendix B.

Put $E_{a,r} = \Lambda_r$ with trivial inertia, and $R_{a,r} = \Lambda_r[[\Gamma_a]]$, the Iwasawa (measure-type) completed group algebra, not an algebra of locally analytic distributions. Use the derived category of pseudocompact modules, computed by completed free resolutions, with completed tensor and continuous derived mapping spectra. Root-level inverse systems are used to present these modules; we do not identify this category with $\varprojlim_n D(\Lambda_r[G_n])$ under derived scalar reduction. In particular the augmentation's finite presentations are not asserted derived-cartesian over finite group rings.

There are two independent root indices in mapping computations: a finite quotient presenting a distribution coefficient, and a possibly larger quotient computing its continuous action. For a pseudocompact target $W = \varprojlim_n W_n$, continuous equivariant Hom is computed by resolving the source and then taking

$$R \varprojlim_n C_{\text{cts}}^*(\Gamma_a \times \Gamma_b, R \text{Hom}_{\Lambda_r}^{\text{cts}}(V, W_n)).$$

The cochains inside are inflation colimits over sufficiently large action levels, not cochains of the single group presenting W_n . This is the completed equivariant bar model for the mapping spectra in the preceding definition. It retains derived continuous maps which a diagonal finite-level Hom calculation would miss.

Lemma A.2 (The source orientation from the completed augmentation). *The augmentation $E_{a,r}$ is perfect over $R_{a,r}$. Its relative dual is*

$$\mathfrak{I}_{a,r} := R \text{Hom}_{R_{a,r}}^{\text{cts}}(E_{a,r}, R_{a,r}) \simeq \begin{cases} \Lambda_r, & \Gamma_a = 0, \\ \Lambda_r(-1)[-1], & \Gamma_a = \Gamma, \end{cases}$$

as a left $R_{a,r}$ -module with augmentation action. Evaluation gives, for every right module M , the canonical natural equivalence

$$M \widehat{\otimes}_{R_{a,r}}^L \mathfrak{I}_{a,r} \xrightarrow{\sim} R \text{Hom}_{R_{a,r}}^{\text{cts}}(E_{a,r}, M). \quad (17)$$

For $M = E_{a,r}$ this identifies the trace factor with the cochain factor, with both augmentation-
endomorphism actions:

$$J_{a,r} \otimes \mathfrak{l}_{a,r} \simeq A_{a,r}, \quad J_{a,r} = E_{a,r} \hat{\otimes}_{R_{a,r}}^L E_{a,r}.$$

When used as a scalar tensor factor we write $\mathfrak{l}_{a,r}$ for the underlying orientation line of $\mathfrak{l}_{a,r}$. The augmentation-
endomorphism/cochain actions are still those transported by the evaluation comparison; the underline does not replace them by trivial actions. In the last formula the tensor is over Λ_r with this underlying line; the augmentation action is retained in the tensor defining J .

Proof. At a node choose the fixed Kummer coordinate and write $T = \gamma - 1$. The two-term completed free resolution $[R_{a,r} \xrightarrow{T} R_{a,r}]$ of the augmentation, in degrees $-1, 0$, is exact since T is a non-zero-divisor even over Λ_r . Dualizing it gives the augmentation in degree 1. Intrinsically its degree-one line is $(I/I^2)^\vee = \Lambda_r(-1)$, so the identification has the stated Tate twist, rather than a basis-dependent scalar shift. Evaluation of a finite free resolution proves (17) for every M and on its full mapping category. Taking $M = E_{a,r}$ gives $A_{a,r}$ by the same augmentation resolution that computes continuous cohomology. Naturality in the source augmentation and the target M proves linearity for the two endomorphism actions. Thus the assertion is a balanced bimodule comparison, not a choice of an isomorphism $DA(-1)[-1] \simeq A$ from graded dimensions. The zero-inertia case is the regular-module evaluation.

One can see the orientation directly in actual continuous root coefficients. For $R_n = \Lambda_r[G_n]$, the continuous action complex is $[R_n \xrightarrow{T} R_n]$ in degrees $0, 1$. Its degree-zero invariant is the norm element; quotient sends it to ℓ times the preceding norm. Its degree-one cokernel is $\Lambda_r(-1)$ and the transition is the identity. At fixed r the former inverse system is pro-zero and the latter is constant, so $R\varprojlim_n R\Gamma_{\text{cts}}(\Gamma, R_n) = \Lambda_r(-1)[-1]$. The two-term complexes identify this with $R\text{Hom}_R(\Lambda_r, R)$, including its maps. In contrast $R\text{Hom}_{\Lambda_r[G_n]}(\Lambda_r, R_n)$ at that single finite group level has no such degree-one class. This exhibits why the continuous-action index cannot be silently tied to the distribution index. $\square$

Definition A.3 (The completed local shriek generating source). For endpoints a, b with the preceding inertia, take actual pseudocompact $(R_{a,r}, R_{b,r})$ -bimodules, retaining their continuous derived mapping spectra. Composition is completed derived tensor over the middle algebra. Its unit is the regular bimodule $R_{a,r}$. The normalized rectangular generator is

$$Q_{ab,r} = \mathfrak{l}_{a,r} \hat{\otimes}_{\Lambda_r} E_{b,r}.$$

Take the smallest full stable idempotent-complete unital convolution source generated by *all* regular diagonal bimodules $R_{a,r}$ and *all* normalized rectangles $Q_{ab,r}$, with perfect scalar coefficients. Inertia graph bimodules are not additional generators. The two endpoints remain in their original order. The shift belongs to the *source* factor, and no shift is inserted in the unit.

Theorem A.4 (Local completed shriek realization). *On this source there is an exact enhanced realization on full kernel mapping categories,*

$$\mathcal{R}_{!,r}^{\text{loc}}(P) = R\text{Hom}_{R_{b,r}}^{\text{cts}}(E_{b,r}, E_{a,r} \hat{\otimes}_{R_{a,r}}^L P), \quad (18)$$

valued in $(A_{a,r}, A_{b,r})$ -bimodules perfect on the right. It sends the completed actual diagonal to the regular unit and the normalized rectangle to the ordinary nearby rectangular bimodule:

$$\mathcal{R}_{!,r}^{\text{loc}}(R_{a,r}) \simeq A_{a,r}, \quad \mathcal{R}_{!,r}^{\text{loc}}(Q_{ab,r}) \simeq A_{a,r} \otimes_{\Lambda_r} A_{b,r}.$$

It has the canonical strong convolution constraint

$$\mathcal{R}_{!,r}^{\text{loc}}(P) \otimes_{A_{b,r}}^L \mathcal{R}_{!,r}^{\text{loc}}(L) \xrightarrow{\sim} \mathcal{R}_{!,r}^{\text{loc}}(P \hat{\otimes}_{R_{b,r}}^L L).$$

The diagonal and rectangle convolution constraints are the continuous assemblies of the actual finite-root shriek operations specified in the proof, not continuous invariants substituted for those operations. The same statements hold for the derived cartesian integral systems generated by these kernels, and after their rationalization: precision reduction is performed on the augmentation resolutions, followed by adic recovery. No finite root level is rationalized first.

Proof. For the actual finite diagonal use the distribution basis of its permutation module $\Lambda_r[G_n]$. The transition $[g] \mapsto [\bar{g}]$ is quotient-fibre summation on distributions. It is the dual of pullback of finite functions, hence is the trace mate for the finite diagonal correspondence. This assertion is on the residual gerbe. Across a normal branch its geometric mate has the exceptional/costalk type described in Proposition B.1; it is not a map to the ordinary pullback of the preceding diagonal kernel. Its inverse limit is $R_{a,r}$ with both regular actions. Unlike the inflation assembly $I_{a,r}$, this is the unit of the completed shriek source. The finite-level identity contractions $\Lambda_r[G_n] \otimes_{\Lambda_r[G_n]} P_n \rightarrow P_n$ commute with these quotient maps, including the contracting homotopies of the regular bar. Thus they give the unit constraints after assembly.

A normalized rectangle has the actual finite-root predecessor $\Lambda_r(-1)[-1] \boxtimes \Lambda_r$ if its source is a node, and the unshifted constant rectangle if the source inertia is zero. Its trace transition is the identity on these coefficient lines. This is a chosen shifted kernel coefficient, not the finite-group shriek image itself. For two rectangles the finite middle shriek operation is the full group-homology bar, with its actual trace maps under quotient, using the geometric input chain in Proposition A.1. Its inverse assembly is $J_{b,r}$ by Proposition A.1. The completed augmentation resolution computes the same $J_{b,r}$, and the comparison of the completed bar with that resolution identifies the evaluation map. Consequently

$$Q_{ab,r} \hat{\otimes}_{R_{b,r}}^L Q_{bc,r} \simeq A_{b,r} \otimes_{\Lambda_r} Q_{ac,r}. \quad (19)$$

The map is precisely (17) on the middle factor. The two outer actions are unchanged. For any fixed finite word of rectangles, the finite computation uses a finite product of full homology bars. Dualizing gives finite products of cohomology bars, whose inflation colimit is the corresponding continuous product bar; each resulting continuous complex is perfect over Λ_r . Dualizing back identifies the trace assembly with the completed tensor computation. No infinite root bar is truncated and no inverse limit is moved through an arbitrary unbounded tensor. Regular factors are handled by their coherent bar contractions. This proves the claimed geometric assembly on the specified local convolution words. For diagrams of their actual continuous maps, retain the independent action and distribution indices described above. Exactness extends an already constructed comparison of such diagrams to its cones and retracts; the ambient realization itself is defined on every continuous kernel map by (18).

The selected coefficient category at a is $\text{thick}_{R_{a,r}}(E_{a,r})$, with its actual continuous module maps. For $Q_{ab,r}$ its action on M is

$$M \hat{\otimes}_{R_{a,r}}^L Q_{ab,r} \simeq R\text{Hom}_{R_{a,r}}^{\text{cts}}(E_{a,r}, M) \otimes E_{b,r}.$$

For M in this thick category the scalar factor is perfect: check $M = E_{a,r}$ by the finite augmentation resolution, then use finite cones and retracts. Thus every generator, and consequently every generated kernel, preserves the selected coefficient types. Formula (18) is therefore its Morita bimodule coordinate, defined by the actual completed kernel action. The

source endomorphisms act through this action and the target endomorphisms by precomposition. Evaluation on the middle augmentation and thick generation prove the strong tensor comparison, exactly as in Proposition 7.1. For a rectangle its value is $A_{a,r} \otimes A_{b,r}$ by (17); for a unit it is $A_{a,r}$ with the regular actions. All composition and unit cells are the ones induced by tensor evaluation and the completed bar, so they are natural on the full mapping categories. No full faithfulness of this realization is required. The unit, orientation and rectangle systems commute with derived precision reduction: their completed free augmentation resolutions reduce to the corresponding resolutions over Λ_r . Their realized images are perfect scalar complexes with uniformly bounded amplitude for each finite expression. Finite cones, convolution and retracts of these cartesian systems preserve the stated comparisons. Their adic limits and subsequent inversion of ℓ therefore give the final assertion; no claim concerns arbitrary precision-dependent infinite convolution words. $\square$

Separate geometric test environment. The next proposition is not an extension of the point-valued source in Definition A.3. Fix the disjoint strata $U = \mathbf{G}_m$ and q . Retain actual sheaf kernels on $U \times U$, sheaves on U with a completed right R_q action for $U \rightarrow q$, sheaves on U with a completed left R_q action for $q \rightarrow U$, and completed bimodules on $q \times q$. We only use the displayed exterior coefficients and two explicitly specified compositions. Through q , contract completed module factors by the actual trace-assembled group bar; through U , use scheme shriek direct image of the sheaf tensor product. Maps of these expressions use those actual tensor, projection and evaluation maps. No common generated enhanced source or strong realization on its full convolution closure is asserted.

Proposition A.5 (Two geometric coarse-passage tests and carrier correction). *At fixed precision put*

$$Q_{Uq} = \Lambda_{r,U} \boxtimes E_q, \quad Q_{qU} = \mathfrak{l}_q \boxtimes \Lambda_{r,U}, \quad Q_{UU} = \Lambda_{r,U} \boxtimes \Lambda_{r,U}, \quad Q_{qq} = \mathfrak{l}_q \otimes_{\Lambda_r} E_q.$$

In the separate test environment just specified, the two compositions are

$$\begin{aligned} Q_{qU} \star Q_{Uq} &\simeq A_q \otimes_{\Lambda_r} Q_{UU}, \\ Q_{Uq} \star Q_{qU} &\simeq R\Gamma_c(U, \Lambda_r) \otimes_{\Lambda_r} Q_{qq}. \end{aligned}$$

The actual unit on U is $\Delta_{U,}\Lambda_r$, not Q_{UU} . There are separate factorwise comparisons with the specified nearby rectangles: target augmentation Hom on Q_{Uq} gives $\Lambda_{r,U} \boxtimes A_q$; source augmentation evaluation on Q_{qU} gives $A_q \boxtimes \Lambda_{r,U}$, with its source cochain action. After the selected adic passage and rationalization these are $H_U \boxtimes A_q$ and $A_q \boxtimes H_U$ through the actual cochain-to-nearby comparisons. These object and action comparisons are not denoted by $\mathcal{R}_!^{\text{loc}}$, whose defined source is point-valued. They identify the two displayed geometric passage factors, not a new common enhanced functor.*

Proof. For $U \rightarrow q \rightarrow U$ the middle trace factor is $J_q \otimes_{\Lambda_r} \mathfrak{l}_q$; its specified augmentation evaluation gives A_q . The two outer sheaves remain on the actual product $U \times U$. For $q \rightarrow U \rightarrow q$, ordinary scheme shriek integration of the middle constant sheaf gives $R\Gamma_c(U, \Lambda_r)$, leaving the outer Q_{qq} . This factor has nonzero cohomology in degrees one and two; replacing U by the zero-inertia point would instead give Λ_r and would change the calculation.

The first factorwise comparison is the actual map

$$R\text{Hom}_{R_q}(E_q, \Lambda_{r,U} \boxtimes E_q) \simeq \Lambda_{r,U} \boxtimes A_q.$$

For its action on $N \in \text{thick}(\Lambda_{r,U})$, it identifies $K_q(R\Gamma_c(U, N) \otimes E_q)$ with $R\Gamma_c(U, N) \otimes A_q$. Here $R\Gamma_c(U, N)$ is scalar perfect in the selected geometric domain, so the tensor–Hom comparison

follows from the finite augmentation resolution. For the reverse rectangle the action on a selected node coefficient M is

$$(M \widehat{\otimes}_{R_q} \mathfrak{l}_q) \boxtimes \Lambda_{r,U} \xrightarrow{\sim} K_q(M) \boxtimes \Lambda_{r,U}.$$

This is precisely (17), retaining the source A_q action. It is the action of the independently specified nearby rectangle $A_q \boxtimes \Lambda_{r,U}$ on $K_q(M)$. These comparisons come from the node factor of the actual exterior kernels; the geometric U factor and its support have not been replaced by point-valued Morita coordinates. Inserting these maps in the two contractions gives the middle factors in the statement. All assertions are natural for the coefficient maps in these expressions. Compatible systems obtained from fixed finite expressions admit the same precision reduction, adic recovery and last-step rationalization as the local calculation. This does not construct a functor on arbitrary mixed convolution words or their full kernel mapping categories.

For comparison with the first paper, a rank- N affine carrier has shriek factor $c_N = \Lambda_r(-N)[-2N]$. Its trace-assembled node output is $J_q \otimes c_N$, whereas the bare-node normalized rectangle uses $J_q \otimes \mathfrak{l}_q$. To compute the latter using that carrier one must use the actual coefficient line

$$\mathfrak{l}_q \otimes c_N^{-1} = \Lambda_r(N-1)[2N-1] \quad (20)$$

on the carrier. The vector-bundle trace, with that line, gives $J_q \otimes c_N \otimes (\mathfrak{l}_q c_N^{-1}) = J_q \otimes \mathfrak{l}_q$. For $N = 1$ the compensating line is $\Lambda_r[1]$, not the bare-node line $\Lambda_r(-1)[-1]$. Projection for this invertible coefficient proves the comparison on maps and pairings. This explains precisely which carrier normalization connects its supported trace with nearby ordinary node integration; an uncompensated carrier trace remains different. $\square$

Proposition A.6 (Inertia graph coefficients in the ambient completed category). *For a stratum map in one of the fixed contractions, let $\varphi : \Gamma_b \rightarrow \Gamma_a$ be its actual neutralized inertia map. The completed transposed graph coefficient is*

$$R_{a,r} R_{a,r} R_{b,r},$$

with right action induced by φ . This is an object of the ambient completed bimodule category, not in general of the source of Definition A.3. Its selected coefficient action is restriction along φ . For this additional object the same coordinate expression is separately defined by

$$M_r(P_\varphi) := R\mathrm{Hom}_{R_{b,r}}(E_{b,r}, E_{a,r} \widehat{\otimes}_{R_{a,r}}^L P_\varphi).$$

It is $A_{b,r}$ with the source action through the actual cochain map $A_{a,r} \rightarrow A_{b,r}$. These comparisons respect composition of the fixed inertia maps and all kernel morphisms. They include the rank-drop case $\Gamma_b = 0$, $\Gamma_a = \Gamma$. This assertion concerns graph restriction actions and their graph-only composition; it does not extend Theorem A.4 to a source generated jointly by graphs and rectangles.

Proof. At a finite quotient the actual graph of $B\varphi$ on the product has the permutation coefficient on the corresponding graph cosets. Viewed as a left/right bimodule this is the regular source group algebra with the right target action through φ . Quotient-fibre trace maps give its completed coefficient as above. Its tensor action is $M \widehat{\otimes}_{R_a} R_a$, with the restricted right R_b -action, so it sends the augmentation to the augmentation and preserves its thick closure. Evaluating $M_r(P_\varphi)$ gives $R\mathrm{Hom}_{R_b}(E_b, E_b) = A_b$. Naturality of the restriction functor on augmentation endomorphisms gives the left action by the actual continuous cochain restriction, not an arbitrary algebra map with the same cohomology. For a composite $\Gamma_c \rightarrow \Gamma_b \rightarrow \Gamma_a$ the kernel comparison is the regular cancellation $R_a \widehat{\otimes}_{R_b} R_b \rightarrow R_a$. Its

tensor associativity and unit maps are already the finite graph composition maps and remain so after completion. This proves the local coefficient faces of the fixed refinement diamond, without proving gluing on the complete exceptional fibres. $\square$

Remark A.7 (Why the ambient graph is an additional object). At $r = 1$, a word from Γ to 0 in the point-valued generating source must contain a rectangle. Cancelling regular units and using (19) leaves a perfect scalar factor times that rectangle. Thus its underlying $\mathbf{F}_\ell$ complex is perfect; finite cones and retracts preserve this property. The rank-drop graph $\mathbf{F}_\ell[[T]]\mathbf{F}_\ell[[T]]\mathbf{F}_\ell$ has infinite-dimensional degree-zero cohomology and is not an object of that Hom category. Adding regular units explicitly does not remove this distinction.

Proposition A.8 (Local evaluation and normalized trace). *The node normalization is compatible with the actual trace-dual pairing. Explicitly, if $\eta : J_q \otimes \mathfrak{l}_q \rightarrow A_q$ is (17), the normalized pairing is*

$$A_q \otimes A_q \xrightarrow{\eta^{-1} \otimes 1} J_q \otimes \mathfrak{l}_q \otimes A_q \xrightarrow{\text{ev} \otimes 1} \mathfrak{l}_q.$$

It is balanced for the two cochain actions and nondegenerate, with their graded side conventions. Before normalization the trace pairing has target Λ_r ; the carrier pairing has target c_N . Tensoring by (20) gives exactly the displayed target $\mathfrak{l}_q$.

Proof. Compute the first evaluation with the finite free augmentation resolution and its dual. The two contraction orders are the same tensor–Hom evaluation: moving an augmentation endomorphism across the contraction precomposes one factor and postcomposes the other. This proves balancing and naturality on the full endomorphism mapping category. The completed bar comparison identifies the other contraction with the dual of inflation, hence with the finite-root trace assembly, as in Proposition A.1. The orientation map is evaluation of that same resolution, so it does not change the pairing by an unspecified scalar. The resulting map pairs degree zero with degree one; those two free lines are perfect duals with value $\Lambda_r(-1)[-1]$. Evaluation and coevaluation obey their usual triangle identities. Proper vector-bundle trace and scalar projection give the last assertion, retaining its twist and shift. These are local pairings on the selected inertia charts; no coefficient duality across an attached whole fibre is inferred. $\square$

A.2 Comparison with the actual three-realization carrier trace

Write $\mathbb{T}_{N,r}(V)$ for the actual trace-assembled rank- N carrier functor of [1], not for a newly defined invariants functor. Its coefficient arrows and trace maps are those of

`lem:frag-point-trace-diagram-descent` there. Write $K_r(M) = R\text{Hom}_R^{\text{cts}}(E_r, M)$ and $J_r(M) = M \hat{\otimes}_R^L E_r$.

Theorem A.9 (The normalized carrier interface on selected coefficients). *Let $M \in \text{thick}_R(E_r)$ and put $d_{N,r} = \mathfrak{l}_r c_N^{-1} = \Lambda_r(N-1)[2N-1]$. There is a natural equivalence of the actual functors*

$$\Theta_{N,M} : \mathbb{T}_{N,r}(M \otimes d_{N,r}) \xrightarrow{\sim} K_r(M). \quad (21)$$

It respects the augmentation-cochain action, evaluation, the actual root trace transitions and coefficient-reduction maps. It is natural on the full selected mapping category, not only for the constant coefficient. It therefore intertwines the carrier comparison with the kernel actions of Theorem A.4, including their maps and composition constraints; at a zero-inertia point use the identity coefficient comparison. This does not assert a common realization for the ambient graph objects or the geometric coarse-passage tests.

Adic recovery and subsequent rationalization concern only compatible systems $\{M_r\}_r$ equipped with specified derived precision-reduction equivalences and the uniformly bounded

scalar-perfect presentations used here, for instance systems constructed by one fixed finite generating expression with compatible maps and retracts. The augmentation comparisons and carrier coefficients use those same reduction maps. A single fixed-precision M , or unrelated choices of M_r , is not asserted to admit such a recovery.

For the fixed node carrier $N = 1$, the coefficient supplied to the first paper's supported realization is $M[1]$. Its actual supported analytic and nearby costalk realizations of $M[1]$ are thereby identified with the ordinary nearby cochain realization of M . This is a comparison of these selected point-valued functors, not an equality of stalk and costalk functors on arbitrary attached objects.

Proof. Every object in $\text{thick}_R(E_r)$ is perfect over Λ_r : this holds for E_r and is preserved by finite cones and retracts. It therefore belongs to the point coefficient domain of the first paper. Its actual trace comparison gives

$$\phi_V : \mathbb{T}_{N,r}(V) \xrightarrow{\sim} D_{\Lambda_r} K_r(D_{\Lambda_r} V) \otimes c_N \simeq J_r(V) \otimes c_N.$$

The last map is evaluation on the same finite free completed augmentation resolution; contragredient actions and the graded side exchanges are retained. In particular this use of the first paper is legitimate for M , but would not be legitimate for the infinite-rank kernel R regarded as a coefficient. Projection for the invertible scalar line $d_{N,r}$ and (17) now give the specified composite

$$\mathbb{T}_{N,r}(M \otimes d_{N,r}) \xrightarrow{\phi} J_r(M) \otimes c_N \otimes d_{N,r} = J_r(M) \otimes \mathbb{I}_r \xrightarrow{\text{ev}} K_r(M).$$

This defines Θ on maps, including its module action. The last action is the augmentation-endomorphism action on the relative dual $R\text{Hom}_R(E_r, R)$; it is not discarded when its underlying orientation line is written as $\Lambda_r(-1)[-1]$. Both pairings are contractions of the augmentation bar with its dual, and the finite-root pairing is precisely the one identified by `lem:frag-point-trace-diagram-descent`. Thus balancing identifies the first paper's transpose cup action with this action. This proves the module assertion and fixes the map, not merely its value as a scalar complex.

For a coefficient map $f : M \rightarrow M'$, the square is explicitly

$$\begin{array}{ccc} \mathbb{T}_{N,r}(M \otimes d_{N,r}) & \xrightarrow{\Theta_{N,M}} & K_r(M) \\ \mathbb{T}_{N,r}(f \otimes 1) \downarrow & & \downarrow K_r(f) \\ \mathbb{T}_{N,r}(M' \otimes d_{N,r}) & \xrightarrow{\Theta_{N,M'}} & K_r(M'). \end{array}$$

It is the square of evaluation and the actual trace functor on mapping categories. The trace transitions in a chosen presentation satisfy $e_n(t_{m,n} \otimes 1) = e_m(1 \otimes \inf_{n,m})$; tensoring by d_N and applying the same evaluation preserves that identity. The coefficient-reduction square is the actual square of the cited lemma, followed by reduction of the finite free augmentation resolution. This proves compatibility with both indices before recovery.

For a kernel P in Definition A.3, its action $F_P(M) = M \hat{\otimes}_R P$ lands in the target selected category. Substitution of $F_P(M)$ into the square above proves compatibility with coefficient maps and kernel maps. For two kernels substitute the actual tensor associativity map $F_L F_P(M) \rightarrow F_{P \otimes L}(M)$; naturality identifies the two carrier comparisons along this map. The same substitution on its higher simplices gives their coherence. At the point this is a proved interface for the local kernel actions, not an application of the first paper to growing diagonal kernels.

Finally `cor:frag-C2-local-closed` of [1] identifies $\mathbb{T}_{N,r}(V)$ with the actual supported analytic and nearby objects, using the same dual star comparison and source twist. Apply

it to $V = M \otimes d_N$. This proves the last assertion for $N = 1$. We have not invoked that paper's separate orientation-factor formula outside its stated trivial-character domain; here the comparison for nontrivial selected M is proved by relative augmentation evaluation. For the compatible systems specified in the statement, these perfect reductions supply integral recovery, followed by rationalization; no lift of an isolated M is inferred. $\square$

A.3 Ordered completion and coupled inertia

Lemma A.10 (The two root indices cannot be replaced by the diagonal). *At fixed r set $R_n = \Lambda_r[G_n]$, with distribution quotient transitions. The correct completed equivariant computation is*

$$R \varprojlim_n R\Gamma_{\text{cts}}(\Gamma, R_n) \simeq \Lambda_r(-1)[-1].$$

Replacing continuous cohomology by the cohomology of the same finite group presenting R_n gives instead a pro-zero system of norm lines, hence zero inverse limit. Thus such a replacement loses a nonzero derived map, not merely an optional coherence choice.

Proof. The completed augmentation resolution gives $[R_n \xrightarrow{T} R_n(-1)]$ in degrees 0, 1, with the coordinate-free interpretation of the twist. Its kernel is the norm line. The quotient map sends the norm at level $n+1$ to ℓ times the norm at level n , so this kernel system is pro-zero at fixed r . Its cokernel is $\Lambda_r(-1)$ with identity transition. These systems are Mittag-Leffler; the inverse limit is the displayed degree-one line.

For the alternative calculation the regular finite module R_n is coinduced from the trivial subgroup of G_n . Hence $R\Gamma(G_n, R_n)$ is its invariant norm line in degree zero, with no positive cohomology. The maps on these norm lines induced by distribution quotient are multiplication by ℓ and have zero derived inverse limit. At a fixed distribution level n , continuous cohomology instead requires inflation over all larger action levels with R_n inflated as coefficient. That additional limit direction supplies the degree-one class. The replacement just tested omits it. In particular the orientation morphisms and product attaching maps cannot be certified by testing only the single-level finite groups. $\square$

Proposition A.11 (Coupled inertia of the completed diagonal). *Fix a node and a precision r . Write $R = \Lambda_r[[\Gamma]]$ and let $R_n = \Lambda_r[G_n]$ carry the left/right action of $\Gamma \times \Gamma$ through $(g, h)[a] = [gah^{-1}]$. There is a specified equivalence*

$$R \varprojlim_n R\Gamma_{\text{cts}}(\Gamma \times \Gamma, R_n) \simeq \mathfrak{l} \otimes_{\Lambda_r} A_r, \quad (22)$$

where the second factor is the cochain algebra of the diagonal stabilizer. The first factor belongs to the quotient $(\Gamma \times \Gamma)/\Delta\Gamma$, identified with the source inertia by $(g, h) \mapsto gh^{-1}$. In particular the two inertia directions do not contribute two independent regular-module orientation lines.

More generally, for an actual stratum map $\varphi : \Gamma_b \rightarrow \Gamma_a$, the completed graph coefficient $P_\varphi = R_a R_a R_b$ satisfies

$$R\Gamma_{\text{cts}}(\Gamma_a \times \Gamma_b, P_\varphi) \simeq \mathfrak{l}_a \otimes A_b.$$

Here cohomology with a completed coefficient means the ordered continuous-action/distribution assembly. Cancellation of the source orientation by augmentation evaluation gives precisely A_b with its actual A_a -action in Proposition A.6. These are ambient equivariant coefficient comparisons. The graph coordinate is the separately defined $M_r(P_\varphi)$, not a claim that the graph belongs to the point-valued generating source. No geometric product attaching map is implicit in them.

Proof. The actual group isomorphism

$$\Gamma_a \times \Gamma_b \longrightarrow \Gamma_a \times \Gamma_b, \quad (g, h) \longmapsto (g\varphi(h)^{-1}, h)$$

identifies the graph subgroup with the second factor and its quotient with the first. In these coordinates the first factor acts regularly on P_φ and the second acts trivially. The continuous bar on this product, or its finite completed augmentation resolution, therefore computes the displayed cohomology as $\mathbb{L}_a \otimes A_b$. The resolution has finite length and finite free terms. Consequently its application to the distribution system commutes with its derived inverse limit; no direct/inverse interchange is involved. At every fixed distribution presentation its cochains still use all dominating action levels.

For the diagonal, a generator gives the two-variable cochain Koszul differential determined by

$$T, \quad (1 + T)^{-1} - 1 = -T(1 + T)^{-1}.$$

The invertible elementary change of these two generators turns this into $T, 0$. The T direction has inverse limit $\mathbb{I}$ by Lemma A.10; the zero direction is the cochain complex of the diagonal stabilizer. Keeping the cotangent lines records its Tate twist and the sign from the inverse right action. This calculation identifies the map coming from the actual group isomorphism, rather than merely its graded dimensions.

Finally, source augmentation evaluation identifies, naturally in any completed bimodule P ,

$$R\mathrm{Hom}_{R_a}(E_a, P) \simeq (E_a \widehat{\otimes}_{R_a}^L P) \otimes_{\Lambda_r} \mathbb{L}_a.$$

Applying target augmentation Hom identifies the preceding coupled calculation with (18), after cancelling this orientation line. The source and target cochain actions are retained throughout by precomposition; for a graph they give the restriction induced by φ . Thus the comparison is the graph realization map already specified, including its bimodule structure. Group composition and the relative bar augmentation give the composition maps, not a separately chosen splitting of the two-term cohomology. $\square$

B An obstruction to ordinary distribution gluing

The ordinary-pullback inverse gluing route is excluded by an actual attaching square. The result does not affect the nearby theorem; it diagnoses a different source construction.

Proposition B.1 (The actual diagonal branch transition and its mate). *Let $\mathcal{X}_n = [D_{z_n}/G_n]$ be a smooth strictly local root branch, $x = z_n^{\ell_n}$, and let $q : \mathcal{X}_m \rightarrow \mathcal{X}_n$, $m > n$, be its actual transition. Put $F_n = \Delta_{\mathcal{X}_n, !}\Lambda_r$. There is a canonical map*

$$\beta_{n,m} : (q \times q)^* F_n \longrightarrow F_m \tag{23}$$

which is the identity on the open diagonal. On the closed atlas fibre it is the function-inflation map

$$\iota_{n,m} : \Lambda_r[G_n] \longrightarrow \Lambda_r[G_m], \quad [a] \longmapsto \sum_{\bar{b}=a} [b].$$

This identifies the coupled diagonal attaching map on this branch, including its degree-zero product-open link face. The maps β compose by the actual adjunction comparison.

Its geometric Verdier mate is an actual trace map

$$(q \times q)_! DF_m \longrightarrow DF_n. \tag{24}$$

Since these branch stacks are smooth of dimension one, $DF_n = F_n(1)[2]$. Thus the mate also gives $(q \times q)!F_m \rightarrow F_n$. The quotient-fibre summation on distributions is the dual of $\iota_{n,m}$ on the corresponding costalk face, with this geometric twist and shift retained. It is not an ordinary-pullback transition on the stalks of F_m .

Proof. The diagonal is finite and representable, so its shriek and star images agree. Adjunction for $\Delta_{\mathcal{X}_m}^*$ and $\Delta_{\mathcal{X}_m,*}$ constructs (23) from

$$q^* \Delta_{\mathcal{X}_n}^* \Delta_{\mathcal{X}_n,*} \Lambda_r \longrightarrow q^* \Lambda_r = \Lambda_r,$$

using the actual diagonal counit. No full faithfulness of the stack diagonal is used. On finite group representations this counit is evaluation at the identity coset in the coinduced permutation module. Its adjoint is pullback of functions along $G_m \rightarrow G_n$, giving the stated $\iota_{n,m}$. The adjunction unit/counit identities identify the maps for three levels.

On the product atlas at level m , the open diagonal has the $|G_m|$ branches $z'_m = \zeta z_m$. Its degree-zero product-open link is therefore $\Lambda_r[G_m]$. The attaching map from F_m is the identity on these branches. For $(q \times q)^* F_n$ it is $\iota_{n,m}$: the label of a level- n branch restricts to all its level- m lifts. Consequently the square for β is

$$\begin{array}{ccc} \Lambda_r[G_n] & \xrightarrow{\iota_{n,m}} & \Lambda_r[G_m] \\ \iota_{n,m} \downarrow & & \downarrow 1 \\ \Lambda_r[G_m] & \xrightarrow{1} & \Lambda_r[G_m]. \end{array}$$

This calculation is on the actual branches, with their permutation actions. Higher link degrees and the connecting arrow are retained in the actual adjunction map, rather than replaced by a splitting of their cohomology.

Apply geometric Verdier duality to this whole map. The identity $D(q \times q)^* = (q \times q)!D$, followed by shriek adjunction, gives (24). Proper duality for the diagonal and the smooth-branch dualizing coefficient give $DF_n = \Delta_{\mathcal{X}_n,*} \Lambda_r(1)[2]$. Dualizing a stalk map gives a costalk map. On these finite free coefficient faces the transpose of $\iota_{n,m}$ is $[b] \mapsto [\bar{b}]$. This proves the assertion about the specified distribution transition and its type. Dualizing the composition identity for β proves trace transitivity in the same diagram, including its boundary mates. $\square$

Corollary B.2 (A stalkwise distribution gluing is impossible). *On this branch there is no morphism $F_m \rightarrow (q \times q)^* F_n$ that is the identity on the open diagonal and is quotient-fibre summation on the closed atlas fibre. More generally there is no such morphism with any map on the closed fibre. Thus a global completed unit cannot be built by gluing the local regular distribution modules using ordinary pullback and unchanged open unit maps.*

Proof. Compatibility with the degree-zero attaching maps would require $\iota_{n,m}c = 1$ on $\Lambda_r[G_m]$, where $c : \Lambda_r[G_m] \rightarrow \Lambda_r[G_n]$ is the putative closed map. Reduction modulo ℓ makes this impossible by dimension, since $|G_n| < |G_m|$. For the proposed quotient map specifically, $\iota_{n,m}c$ sends a basis vector to the sum of all vectors in its quotient fibre; it is not the identity, even before reduction. This obstruction is already visible in the degree-zero face and cannot be repaired by a higher homotopy of the same maps. $\square$

The corollary does not rule out a completed root shriek interface: (24) is its actual alternative transition. It does rule out identifying that exceptional transition with an ordinary stalkwise inverse system. In particular the local pseudocompact module source does not itself supply the global sheaf or costalk descent category. On a whole singular fibre, one must retain its actual dualizing complex in this construction; the smooth-branch formula (1)[2] cannot be used as the dualizing complex of a nodal union.

Lemma B.3 (The singular-support dualizing correction is not a line). *On an ordinary strictly local nodal curve C with two smooth branches, the stalk of $\omega_C = D_C \Lambda_r$ at the node i has*

$$H^{-2}(i^* \omega_C) = \Lambda_r(1)^{\oplus 2}, \quad H^{-1}(i^* \omega_C) = \Lambda_r,$$

and no other cohomology. In particular ω_C is not an invertible shifted coefficient line. The same local test applies on a reduced nodal atlas, with the actual stabilizer action retained.

Proof. For the punctured inclusion j , its two disjoint branches give $H^0(i^* Rj_* \Lambda_r) = \Lambda_r^{\oplus 2}$ and $H^1(i^* Rj_* \Lambda_r) = \Lambda_r(-1)^{\oplus 2}$. The restriction of the constant sheaf in degree zero is the diagonal map. The localization fibre sequence

$$i^! \Lambda_r \longrightarrow \Lambda_r \longrightarrow i^* Rj_* \Lambda_r$$

therefore has costalk cohomology Λ_r in degree one and $\Lambda_r(-1)^{\oplus 2}$ in degree two. These modules are free over Λ_r . Point Verdier duality identifies $i^* \omega_C = D_{\Lambda_r} i^! \Lambda_r$, yielding the stated cohomology. An invertible complex over a local coefficient ring has one rank-one cohomology line after reduction to the residue field, which this stalk does not. This is a necessary local test on an atlas as well; forgetting its stabilizer action cannot turn an invertible equivariant line into the displayed complex. $\square$

The obstruction excludes ordinary stalk/pullback distribution completion, not every root realization. The actual mate retains $\Delta_* \omega_X$, and the preceding calculation prevents replacing its coefficient by one invertible line on a nodal support. Constructing a kernel calculus with compatible units, transitions and convolution is a separate problem. No such global root source or fourth main theorem is asserted here. The independent research draft *Exceptional and Completed Root Kernel Calculus: A Single-Transition Problem* records that problem and the auxiliary finite diagrams. Its first test concerns one complete nodal support, one actual transition and one crossing rectangle; the four-model coherence problem depends on that test.

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References

- [1] Y. Jiang, *Local Kummer Realizations and Verdier Duality*, version dated 22 September 2026 (fixed companion input version).
- [2] Y. Jiang, *Morita Coefficients, Incidence, and Semistable Refinement*, version dated 20 September 2026 (fixed companion input version).
- [3] Y. Liu and W. Zheng, *Enhanced six operations and base change theorem for higher Artin stacks*, arXiv:1211.5948.

- [4] Y. Liu and W. Zheng, *Enhanced adic formalism and perverse t -structures for higher Artin stacks*, arXiv:1404.1128.
- [5] B. Toën, *The homotopy theory of dg-categories and derived Morita theory*, [arXiv:math/0408337v7](#), 19 October 2006.
- [6] D. Ben-Zvi, J. Francis and D. Nadler, *Integral Transforms and Drinfeld Centers in Derived Algebraic Geometry*, J. Amer. Math. Soc. **23** (2010), no. 4, 909–966; [arXiv:0805.0157v5](#), 4 March 2010.